

Modern Practical Control Systems

Chapter 1

System Modeling with Ideal Components

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Chapter 1

System Modeling with Ideal Components

Learning Objectives

After completing this chapter, you will be able to:

- Understand the purpose and process of mathematical modeling
- Apply ideal sensor and actuator assumptions appropriately
- Derive differential equations from physical principles
- Convert between different model representations
- Recognize analogies between different physical domains

The foundation of control systems engineering is the mathematical model—a set of equations that describe how a system behaves. Before we can analyze stability, design controllers, or implement feedback, we must first capture the essential dynamics of the system we wish to control.

This chapter introduces systematic approaches to modeling physical systems across diverse engineering domains. We begin with ideal assumptions: perfect sensors that measure exactly what we want, and ideal actuators that respond instantly with unlimited capability. These assumptions let us focus on the fundamental physics before adding real-world complications in later chapters.

1.1 The Art and Science of Modeling

1.1.1 Why Model?

Mathematical models serve several critical purposes in control engineering:

1. **Prediction:** Models let us simulate system behavior before building hardware, saving time and money.
2. **Analysis:** We can mathematically determine stability, performance limits, and sensitivities.
3. **Design:** Controllers are designed based on models—better models lead to better controllers.
4. **Understanding:** The modeling process itself reveals what physical parameters matter most.

“All models are wrong, but some are useful.” — George E.P. Box

A model is always an approximation. The goal is not perfect fidelity but rather capturing the dynamics relevant to control at the frequencies and amplitudes of interest.

1.1.2 The Modeling Process

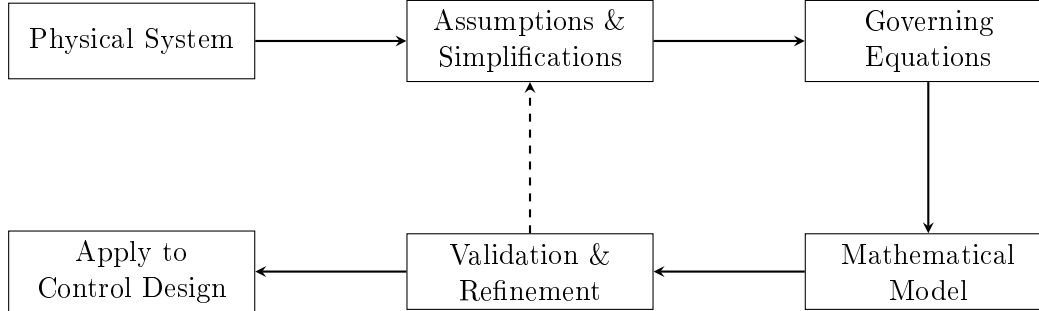


Figure 1.1: The iterative modeling process.

The modeling process typically involves:

1. **Define the system boundary:** What is included? What is external?
2. **Identify inputs and outputs:** What can we manipulate? What do we measure?
3. **List assumptions:** What simplifications are appropriate?
4. **Apply physical laws:** Conservation of energy, mass, momentum, charge, etc.
5. **Derive equations:** Combine physical laws into mathematical form.
6. **Validate:** Compare model predictions with experimental data.
7. **Refine:** Add complexity only where needed.

1.2 Ideal Assumptions

Throughout this chapter, we make two fundamental idealizations:

1.2.1 Ideal Sensors

An **ideal sensor** provides a perfect measurement of the quantity of interest:

- **No delay:** The measurement is instantaneous
- **No noise:** The measurement has no random errors
- **No bias:** The measurement has no systematic offset
- **Infinite bandwidth:** All frequencies are captured equally
- **Infinite resolution:** Any value can be measured exactly
- **No loading:** The sensor does not affect the measured quantity

Mathematically, if $x(t)$ is the true value, the ideal sensor output is:

$$y_{\text{measured}}(t) = x(t) \quad (1.1)$$

1.2.2 Ideal Actuators

An **ideal actuator** produces exactly the commanded output:

- **No delay:** Response is instantaneous
- **No dynamics:** No transient behavior in the actuator itself
- **Infinite bandwidth:** Can track any commanded signal perfectly
- **Unlimited force/torque/power:** No saturation limits
- **Infinite acceleration:** Can change output instantly
- **No backlash or friction:** Perfect mechanical transmission

Mathematically, if $u(t)$ is the commanded input, the ideal actuator produces:

$$F_{\text{actual}}(t) = u(t) \quad (\text{or voltage, torque, flow rate, etc.}) \quad (1.2)$$

Important Note

Real sensors and actuators violate all of these assumptions to varying degrees. Chapters on sensor modeling and actuator dynamics will address these non-idealities. For now, ideal assumptions let us focus on the plant dynamics.

1.3 Fundamental Physical Laws

Most dynamic systems can be modeled using a small set of conservation principles:

1.3.1 Newton's Laws (Mechanical)

Newton's Second Law for translation:

$$\sum F = ma = m \frac{d^2x}{dt^2} \quad (1.3)$$

Rotational analog:

$$\sum \tau = J\alpha = J \frac{d^2\theta}{dt^2} \quad (1.4)$$

where J is the moment of inertia and α is angular acceleration.

1.3.2 Kirchhoff's Laws (Electrical)

Kirchhoff's Voltage Law (KVL): The sum of voltages around any closed loop is zero:

$$\sum_{\text{loop}} V = 0 \quad (1.5)$$

Kirchhoff's Current Law (KCL): The sum of currents at any node is zero:

$$\sum_{\text{node}} i = 0 \quad (1.6)$$

1.3.3 Conservation of Energy (Thermal)

$$\frac{dE}{dt} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} + \dot{W} \quad (1.7)$$

For thermal systems:

$$mC_p \frac{dT}{dt} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \quad (1.8)$$

1.3.4 Conservation of Mass (Fluid/Chemical)

$$\frac{dm}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \quad (1.9)$$

For incompressible fluids with density ρ and volume V :

$$\rho \frac{dV}{dt} = \rho Q_{\text{in}} - \rho Q_{\text{out}} \quad (1.10)$$

where Q is volumetric flow rate.

1.4 Model Representations

The same dynamic system can be represented in several equivalent forms:

1.4.1 Differential Equation Form

The most direct form, derived from physical laws:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_m \frac{d^m u}{dt^m} + \cdots + b_0 u \quad (1.11)$$

1.4.2 Transfer Function Form

Taking the Laplace transform (with zero initial conditions):

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0} \quad (1.12)$$

1.4.3 State-Space Form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (1.13)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}u \quad (1.14)$$

where $\mathbf{x}$ is the state vector containing the minimum information needed to predict future behavior.

1.4.4 Block Diagram Form

A graphical representation showing signal flow and interconnections.

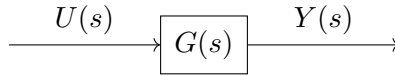


Figure 1.2: Simple block diagram representation.

Concept	Mechanical (Trans.)	Electrical	Thermal
Effort (across)	Force F	Voltage V	Temperature T
Flow (through)	Velocity v	Current i	Heat flow $\dot{Q}$
Resistance	Damper b	Resistor R	Thermal res. R_t
Capacitance	Spring $1/k$	Capacitor C	Thermal cap. C_t
Inductance	Mass m	Inductor L	—

Table 1.1: Analogous quantities across physical domains (force-voltage analogy).

1.5 System Analogies

Remarkably, systems from different physical domains often share the same mathematical structure. This allows us to transfer intuition and analysis techniques across domains.

Example 1.1: Mass-Spring-Damper $\equiv$ RLC Circuit

The mechanical system:

$$m\ddot{x} + b\dot{x} + kx = F(t) \quad (1.15)$$

is mathematically identical to the series RLC circuit:

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = V(t) \quad (1.16)$$

Both have the same transfer function form:

$$G(s) = \frac{1}{ms^2 + bs + k} \equiv \frac{1}{Ls^2 + Rs + 1/C} \quad (1.17)$$

1.6 Standard Test Inputs

To characterize system behavior, we use standard test inputs:

1.6.1 Impulse Input

$$u(t) = \delta(t), \quad U(s) = 1 \quad (1.18)$$

The impulse response $h(t) = \mathcal{L}^{-1}\{G(s)\}$ completely characterizes an LTI system.

1.6.2 Step Input

$$u(t) = A \cdot u(t), \quad U(s) = \frac{A}{s} \quad (1.19)$$

Step response reveals settling time, overshoot, and steady-state error.

1.6.3 Ramp Input

$$u(t) = At \cdot u(t), \quad U(s) = \frac{A}{s^2} \quad (1.20)$$

Ramp response reveals tracking ability for constantly changing references.

1.6.4 Sinusoidal Input

$$u(t) = A \sin(\omega t), \quad U(s) = \frac{A\omega}{s^2 + \omega^2} \quad (1.21)$$

Sinusoidal response at various frequencies reveals the frequency response.

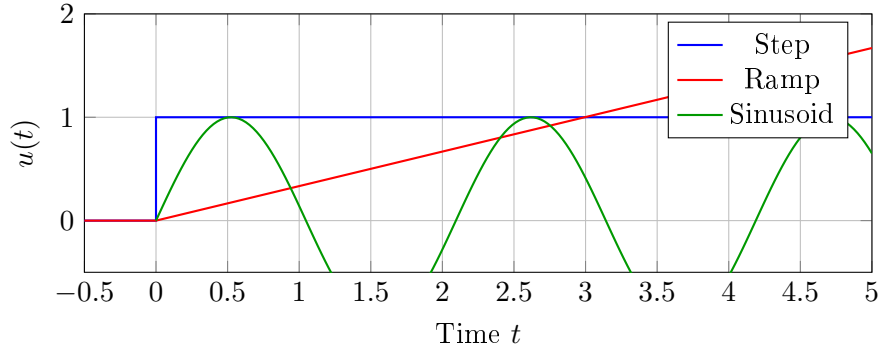


Figure 1.3: Standard test inputs for system characterization.

1.7 First and Second Order System Responses

1.7.1 First-Order Systems

The standard first-order transfer function:

$$G(s) = \frac{K}{\tau s + 1} \quad (1.22)$$

where:

- K = DC gain (steady-state gain)
- τ = time constant

Step Response:

$$y(t) = K \left(1 - e^{-t/\tau} \right) \quad (1.23)$$

Key characteristics:

- At $t = \tau$: reaches 63.2% of final value
- At $t = 4\tau$: reaches 98.2% of final value (settling time)
- Initial slope: K/τ

1.7.2 Second-Order Systems

The standard second-order transfer function:

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (1.24)$$

where:

- K = DC gain
- ω_n = natural frequency (rad/s)
- ζ = damping ratio

Poles:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad (1.25)$$

Step Response (underdamped, $0 < \zeta < 1$):

$$y(t) = K \left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t + \phi) \right] \quad (1.26)$$

where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ and $\phi = \arccos(\zeta)$.

Time-Domain Specifications:

$$\text{Rise time (10-90\%): } t_r \approx \frac{1.8}{\omega_n} \quad (1.27)$$

$$\text{Peak time: } t_p = \frac{\pi}{\omega_d} \quad (1.28)$$

$$\text{Percent overshoot: } \%OS = 100 \cdot e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad (1.29)$$

$$\text{Settling time (2\%): } t_s \approx \frac{4}{\zeta\omega_n} \quad (1.30)$$

1.8 Chapter Organization

The remainder of this chapter presents detailed modeling examples across engineering domains:

- **Section 1B:** Mechanical Systems—translational and rotational dynamics, mass-spring-damper systems, vehicle suspensions, multi-body systems
- **Section 1C:** Electrical Systems—passive circuits, operational amplifiers, filters, power electronics
- **Section 1D:** Electromechanical Systems—DC motors, generators, voice coils, solenoids
- **Section 1E:** Thermal and Fluid Systems—heat transfer, hydraulics, pneumatics, coupled thermal-fluid
- **Section 1F:** Chemical Systems—reaction kinetics, mixing, continuous stirred-tank reactors
- **Section 1G:** Aerospace Systems—aircraft dynamics, spacecraft attitude, rocket propulsion

- **Section 1H:** Biomedical Systems—pharmacokinetics, physiological models, medical devices
- **Section 1I:** Industrial and Computer Systems—manufacturing, queueing, computing

Each section follows a consistent pattern: physical description, governing equations, transfer function derivation, example analysis with step/sinusoidal inputs, and practical considerations.

1.9 Mechanical Systems

Mechanical systems form the intuitive foundation for understanding dynamics. The physical concepts of mass, springs, and dampers have direct analogs in other domains and provide excellent intuition for system behavior.

1.9.1 Translational Mechanical Elements

Mass (Inertia)

Mass resists changes in velocity according to Newton's second law:

$$F = m \frac{dv}{dt} = m \frac{d^2x}{dt^2} = m\ddot{x} \quad (1.31)$$

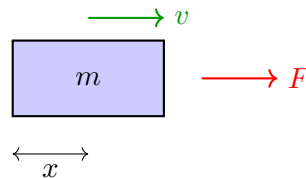


Figure 1.4: Mass element with applied force.

In the Laplace domain:

$$F(s) = ms^2X(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{ms^2} \quad (1.32)$$

Spring (Stiffness)

A linear spring produces a force proportional to displacement:

$$F = kx \quad \text{or} \quad F = k(x_1 - x_2) \quad (1.33)$$

where k is the spring constant (N/m or lb/in).

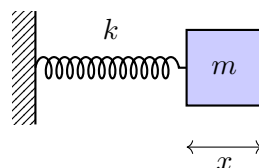


Figure 1.5: Spring element connecting wall to mass.

In the Laplace domain:

$$F(s) = kX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{k} \quad (1.34)$$

Damper (Viscous Friction)

A viscous damper produces a force proportional to velocity:

$$F = b\dot{x} = b \frac{dx}{dt} \quad \text{or} \quad F = b(\dot{x}_1 - \dot{x}_2) \quad (1.35)$$

where b is the damping coefficient (N·s/m or lb·s/in).

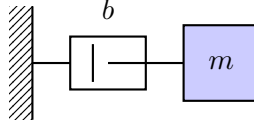


Figure 1.6: Viscous damper element.

In the Laplace domain:

$$F(s) = bsX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{bs} \quad (1.36)$$

1.9.2 Mass-Spring-Damper System

The canonical mechanical system combines all three elements.

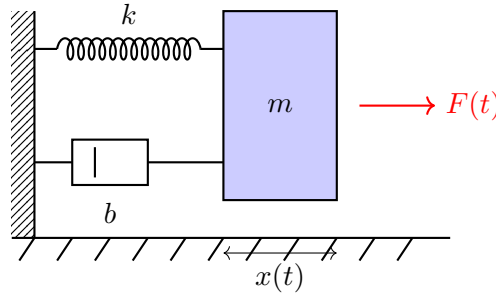


Figure 1.7: Mass-spring-damper system.

Equation of Motion

Applying Newton's second law:

$$m\ddot{x} = F(t) - kx - b\dot{x} \quad (1.37)$$

Rearranging:

$$\boxed{m\ddot{x} + b\dot{x} + kx = F(t)} \quad (1.38)$$

Transfer Function

Taking the Laplace transform with zero initial conditions:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \quad (1.39)$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} \quad (1.40)$$

In standard second-order form:

$$G(s) = \frac{1/k}{(m/k)s^2 + (b/k)s + 1} = \frac{1/k}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1} \quad (1.41)$$

where:

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{natural frequency}) \quad (1.42)$$

$$\zeta = \frac{b}{2\sqrt{km}} \quad (\text{damping ratio}) \quad (1.43)$$

Example 1.2: Automobile Shock Absorber Test

A shock absorber test rig has $m = 50$ kg, $k = 20,000$ N/m, and $b = 1,000$ N·s/m. Find:

- (a) Natural frequency and damping ratio
- (b) The response to a 100 N step force

Solution:

(a) Natural frequency and damping ratio:

$$\omega_n = \sqrt{\frac{20,000}{50}} = \sqrt{400} = 20 \text{ rad/s} = 3.18 \text{ Hz} \quad (1.44)$$

$$\zeta = \frac{1,000}{2\sqrt{20,000 \times 50}} = \frac{1,000}{2,000} = 0.5 \quad (1.45)$$

The system is underdamped.

(b) Transfer function:

$$G(s) = \frac{1}{50s^2 + 1000s + 20000} = \frac{0.00005}{s^2 + 20s + 400} \quad (1.46)$$

For a 100 N step: $Y(s) = \frac{100}{s} \cdot G(s)$

Using partial fractions and inverse Laplace:

$$x(t) = \frac{100}{20000} \left[1 - \frac{e^{-10t}}{\sqrt{1 - 0.25}} \sin(17.32t + 1.047) \right] \quad (1.47)$$

Steady-state displacement: $x_{ss} = \frac{F}{k} = \frac{100}{20000} = 5 \text{ mm}$

1.9.3 Vehicle Suspension System

A quarter-car model represents one corner of a vehicle suspension.

Equations of Motion

For the sprung mass (body):

$$m_s \ddot{x}_s = -k_s(x_s - x_u) - b_s(\dot{x}_s - \dot{x}_u) \quad (1.48)$$

For the unsprung mass (wheel):

$$m_u \ddot{x}_u = k_s(x_s - x_u) + b_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r) \quad (1.49)$$

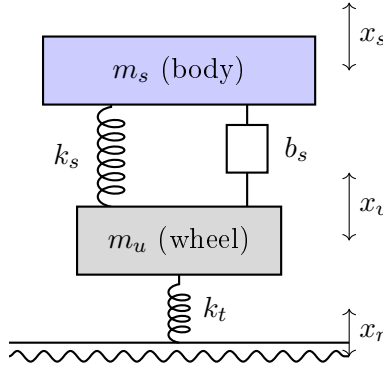


Figure 1.8: Quarter-car suspension model.

State-Space Representation

Defining state variables: $\mathbf{x} = [x_s - x_u, \dot{x}_s, x_u - x_r, \dot{x}_u]^\top$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -k_s/m_s & -b_s/m_s & 0 & b_s/m_s \\ 0 & 0 & 0 & 1 \\ k_s/m_u & b_s/m_u & -k_t/m_u & -b_s/m_u \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \dot{x}_r \quad (1.50)$$

Transfer Functions

From road input $X_r(s)$ to body displacement $X_s(s)$:

$$\frac{X_s(s)}{X_r(s)} = \frac{(b_s s + k_s)k_t}{m_s m_u s^4 + (m_s + m_u)b_s s^3 + [m_s k_t + (m_s + m_u)k_s]s^2 + b_s k_t s + k_s k_t} \quad (1.51)$$

Example 1.3: Quarter-Car Parameters

Typical values for a passenger car:

- Sprung mass: $m_s = 250$ kg (quarter of body mass)
- Unsprung mass: $m_u = 35$ kg (wheel, brake, suspension links)
- Suspension stiffness: $k_s = 16,000$ N/m
- Suspension damping: $b_s = 1,000$ N·s/m
- Tire stiffness: $k_t = 160,000$ N/m

The body bounce frequency: $f_s = \frac{1}{2\pi} \sqrt{\frac{k_s}{m_s}} = 1.27$ Hz

The wheel hop frequency: $f_u = \frac{1}{2\pi} \sqrt{\frac{k_t}{m_u}} = 10.8$ Hz

1.9.4 Rotational Mechanical Systems

Rotational systems follow analogous laws with angular quantities.

Translational	Rotational	Unit
Displacement x	Angle θ	rad
Velocity $v = \dot{x}$	Angular velocity $\omega = \dot{\theta}$	rad/s
Acceleration $a = \ddot{x}$	Angular acceleration $\alpha = \ddot{\theta}$	rad/s ²
Force F	Torque τ	N·m
Mass m	Moment of inertia J	kg·m ²
Spring constant k	Torsional stiffness κ	N·m/rad
Damping b	Rotational damping B	N·m·s/rad

Table 1.2: Translational-rotational analogies.

Rotational Inertia

$$\tau = J\ddot{\theta} = J\dot{\omega} \quad (1.52)$$

Torsional Spring

$$\tau = \kappa\theta \quad \text{or} \quad \tau = \kappa(\theta_1 - \theta_2) \quad (1.53)$$

Rotational Damper

$$\tau = B\dot{\theta} = B\omega \quad (1.54)$$

Example 1.4: Rotating Shaft System

A motor drives a load through a flexible shaft. The motor inertia is $J_m = 0.01 \text{ kg}\cdot\text{m}^2$, load inertia is $J_L = 0.1 \text{ kg}\cdot\text{m}^2$, shaft stiffness is $\kappa = 100 \text{ N}\cdot\text{m}/\text{rad}$, and bearing friction is $B = 0.5 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$.

Equations of motion:

$$J_m\ddot{\theta}_m = \tau_m - \kappa(\theta_m - \theta_L) \quad (1.55)$$

$$J_L\ddot{\theta}_L = \kappa(\theta_m - \theta_L) - B\dot{\theta}_L \quad (1.56)$$

Transfer function from motor torque to load angle:

$$\frac{\Theta_L(s)}{\tau_m(s)} = \frac{\kappa}{J_m J_L s^4 + J_m B s^3 + (J_m + J_L)\kappa s^2 + B\kappa s} \quad (1.57)$$

1.9.5 Gear Trains

Gears transform speed and torque between shafts.

Gear Ratio:

$$n = \frac{N_2}{N_1} = \frac{r_2}{r_1} = \frac{\omega_1}{\omega_2} = \frac{\tau_2}{\tau_1} \quad (1.58)$$

Reflected Inertia: When reflecting load inertia J_L to the motor side:

$$J_{\text{reflected}} = \frac{J_L}{n^2} \quad (1.59)$$

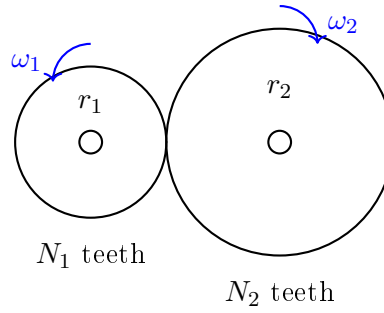


Figure 1.10: Gear pair.

Reflected Damping:

$$B_{\text{reflected}} = \frac{B_L}{n^2} \quad (1.60)$$

Gear Ratio Selection

The gear ratio affects both speed and torque:

- High gear ratio ($n > 1$): Reduces speed, increases torque, reduces reflected inertia
- Low gear ratio ($n < 1$): Increases speed, reduces torque, increases reflected inertia

Optimal gear ratio often maximizes acceleration: $n_{\text{opt}} = \sqrt{J_L/J_m}$

1.9.6 Multi-Body Mechanical Systems**Example 1.5: Two-Mass System**

Consider two masses connected by springs and dampers:

Equations of motion:

$$m_1 \ddot{x}_1 = -k_1 x_1 - b_1 \dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) \quad (1.61)$$

$$m_2 \ddot{x}_2 = F - k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) \quad (1.62)$$

State-space form with $\mathbf{x} = [x_1, \dot{x}_1, x_2, \dot{x}_2]^\top$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_1 + k_2)/m_1 & -(b_1 + b_2)/m_1 & k_2/m_1 & b_2/m_1 \\ 0 & 0 & 0 & 1 \\ k_2/m_2 & b_2/m_2 & -k_2/m_2 & -b_2/m_2 \end{bmatrix} \quad (1.63)$$

1.9.7 Pendulum Systems**Simple Pendulum**

The equation of motion (nonlinear):

$$mL^2 \ddot{\theta} + mgL \sin \theta = \tau \quad (1.64)$$

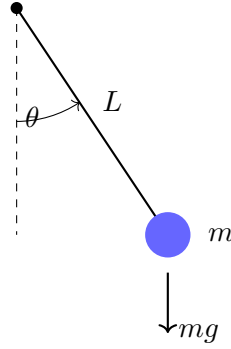


Figure 1.12: Simple pendulum.

For small angles ($\sin \theta \approx \theta$), the linearized equation:

$$\ddot{\theta} + \frac{g}{L}\theta = \frac{\tau}{mL^2} \quad (1.65)$$

Natural frequency: $\omega_n = \sqrt{g/L}$

Inverted Pendulum on Cart

The classic control benchmark problem:

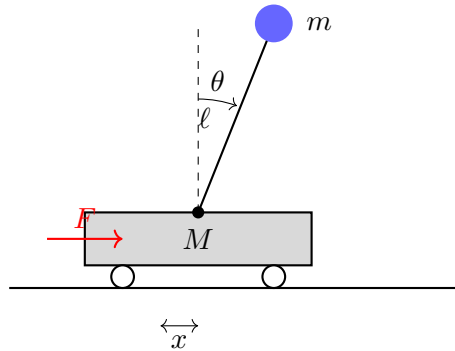


Figure 1.13: Inverted pendulum on cart.

Linearized equations about the upright position ($\theta = 0$):

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F \quad (1.66)$$

$$m\ell\ddot{x} + m\ell^2\ddot{\theta} - mg\ell\theta = 0 \quad (1.67)$$

Transfer function from force to angle:

$$\frac{\Theta(s)}{F(s)} = \frac{1}{(M\ell)s^2 - (M + m)g} \quad (1.68)$$

This system is **open-loop unstable** (pole in the right half-plane) and requires feedback control for stabilization.

1.9.8 Exercises: Mechanical Systems

Exercise 1.1. A spring-mass system has $m = 2$ kg and $k = 200$ N/m. Find:

- (a) The natural frequency in rad/s and Hz
- (b) The transfer function $X(s)/F(s)$
- (c) The response to a 10 N step force

Exercise 1.2. Add a damper with $b = 8$ N·s/m to the system in Exercise 1.

- (a) Calculate the damping ratio
- (b) Find the damped natural frequency
- (c) Determine if the system is underdamped, critically damped, or overdamped

Exercise 1.3. A motor with $J_m = 0.005$ kg·m² drives a load with $J_L = 0.5$ kg·m² through a 10:1 gear ratio.

- (a) Calculate the reflected load inertia at the motor
- (b) Find the total effective inertia seen by the motor
- (c) Derive the transfer function from motor torque to load angle

Exercise 1.4. For the quarter-car model, derive the transfer function from road velocity $\dot{X}_r(s)$ to body acceleration $s^2 X_s(s)$. This represents the acceleration felt by passengers.

Exercise 1.5. Linearize the simple pendulum equation about $\theta_0 = \pi/6$ (30 degrees). Find the equilibrium torque and the linearized dynamics.

1.10 Electrical Systems

Electrical circuits provide clean examples of dynamic systems with well-defined mathematical models. The concepts developed here—impedance, frequency response, filtering—form the historical foundation of control theory.

1.10.1 Passive Circuit Elements

Resistor

Ohm's law relates voltage and current:

$$v(t) = Ri(t) \tag{1.69}$$

In the Laplace domain: $V(s) = RI(s)$

Impedance: $Z_R(s) = R$

Figure 1.14: Resistor element.

Capacitor

The capacitor stores charge:

$$i(t) = C \frac{dv}{dt} \quad \Leftrightarrow \quad v(t) = \frac{1}{C} \int i(\tau) d\tau \quad (1.70)$$

In the Laplace domain: $I(s) = CsV(s)$ or $V(s) = \frac{1}{Cs}I(s)$

Impedance: $Z_C(s) = \frac{1}{Cs}$

Figure 1.15: Capacitor element.

Inductor

The inductor stores magnetic energy:

$$v(t) = L \frac{di}{dt} \quad \Leftrightarrow \quad i(t) = \frac{1}{L} \int v(\tau) d\tau \quad (1.71)$$

In the Laplace domain: $V(s) = LsI(s)$ or $I(s) = \frac{1}{Ls}V(s)$

Impedance: $Z_L(s) = Ls$

Figure 1.16: Inductor element.

1.10.2 RC Circuit (First-Order System)

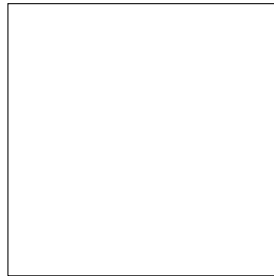


Figure 1.17: RC low-pass filter circuit.

Time-Domain Analysis

Applying KVL:

$$v_{\text{in}} = v_R + v_{\text{out}} = Ri + v_{\text{out}} \quad (1.72)$$

Since $i = C \frac{dv_{\text{out}}}{dt}$:

$$\boxed{RC \frac{dv_{\text{out}}}{dt} + v_{\text{out}} = v_{\text{in}}} \quad (1.73)$$

This is a first-order ODE with time constant $\tau = RC$.

Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} \quad (1.74)$$

Frequency Response

Substituting $s = j\omega$:

$$G(j\omega) = \frac{1}{1 + j\omega RC} \quad (1.75)$$

Magnitude and phase:

$$|G(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad (1.76)$$

$$\angle G(j\omega) = -\arctan(\omega RC) \quad (1.77)$$

Cutoff frequency (where magnitude is $1/\sqrt{2}$ or -3 dB):

$$\omega_c = \frac{1}{RC} = \frac{1}{\tau} \quad (1.78)$$

Example 1.6: RC Circuit Step Response

An RC circuit has $R = 10 \text{ k}\Omega$ and $C = 1 \text{ }\mu\text{F}$. Find:

- (a) Time constant
- (b) Cutoff frequency
- (c) Response to a 5V step input

Solution:

(a) Time constant:

$$\tau = RC = 10,000 \times 10^{-6} = 0.01 \text{ s} = 10 \text{ ms} \quad (1.79)$$

(b) Cutoff frequency:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 0.01} = 15.9 \text{ Hz} \quad (1.80)$$

(c) Step response:

$$v_{\text{out}}(t) = 5(1 - e^{-t/0.01}) = 5(1 - e^{-100t}) \text{ V} \quad (1.81)$$

At $t = \tau = 10 \text{ ms}$: $v_{\text{out}} = 5(1 - e^{-1}) = 3.16 \text{ V}$ (63.2% of final)

1.10.3 RL Circuit

For current as output:

$$L \frac{di}{dt} + Ri = v_{\text{in}} \quad (1.82)$$

Transfer function (voltage to current):

$$\frac{I(s)}{V_{\text{in}}(s)} = \frac{1}{Ls + R} = \frac{1/R}{(L/R)s + 1} \quad (1.83)$$

Time constant: $\tau = L/R$

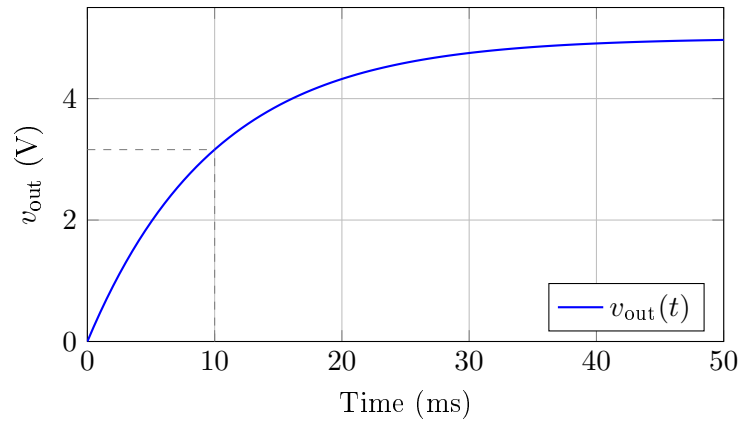


Figure 1.18: RC circuit step response.

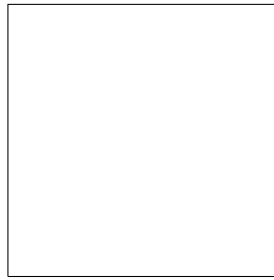


Figure 1.19: RL circuit.

1.10.4 RLC Circuit (Second-Order System)

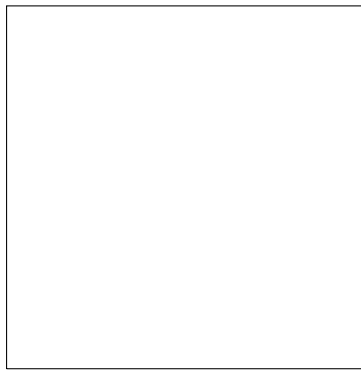


Figure 1.20: Series RLC circuit.

Differential Equation

Applying KVL:

$$v_{\text{in}} = v_R + v_L + v_C = Ri + L \frac{di}{dt} + v_{\text{out}} \quad (1.84)$$

Since $i = C \frac{dv_{\text{out}}}{dt}$:

$$LC \frac{d^2 v_{\text{out}}}{dt^2} + RC \frac{dv_{\text{out}}}{dt} + v_{\text{out}} = v_{\text{in}} \quad (1.85)$$

Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{LCs^2 + RCs + 1} \quad (1.86)$$

In standard form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (1.87)$$

where:

$$\omega_n = \frac{1}{\sqrt{LC}} \quad (\text{resonant frequency}) \quad (1.88)$$

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}} \quad (\text{damping ratio}) \quad (1.89)$$

Quality Factor

The quality factor Q measures the sharpness of resonance:

$$Q = \frac{1}{2\zeta} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad (1.90)$$

High Q means sharp resonance peak and low damping.

Example 1.7: RLC Circuit Design

Design an RLC circuit with resonant frequency $f_n = 1$ kHz and $Q = 10$.

Solution:

Choose $C = 0.1 \mu\text{F}$. Then:

$$L = \frac{1}{(2\pi f_n)^2 C} = \frac{1}{(2\pi \times 1000)^2 \times 10^{-7}} = 0.253 \text{ H} \quad (1.91)$$

For $Q = 10$:

$$R = \frac{1}{Q} \sqrt{\frac{L}{C}} = \frac{1}{10} \sqrt{\frac{0.253}{10^{-7}}} = 159 \Omega \quad (1.92)$$

Damping ratio: $\zeta = 1/(2Q) = 0.05$ (highly underdamped)

1.10.5 Operational Amplifier Circuits

Operational amplifiers (op-amps) are fundamental building blocks for analog control systems.

Ideal Op-Amp Assumptions

- Infinite open-loop gain: $A \rightarrow \infty$
- Infinite input impedance: $Z_{\text{in}} \rightarrow \infty$ (no input current)
- Zero output impedance: $Z_{\text{out}} = 0$
- Infinite bandwidth

With negative feedback, these lead to the “golden rules”:

1. $V_+ = V_-$ (virtual short)
2. $I_+ = I_- = 0$ (no input current)

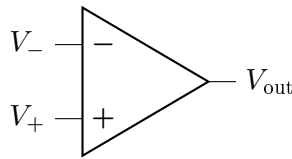


Figure 1.21: Op-amp symbol.

Inverting Amplifier

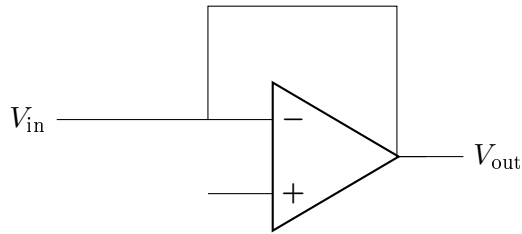


Figure 1.22: Inverting amplifier.

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1} \quad (1.93)$$

Non-Inverting Amplifier

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1} \quad (1.94)$$

Integrator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -\frac{1}{RCs} \quad (1.95)$$

Time-domain:

$$v_{\text{out}}(t) = -\frac{1}{RC} \int_0^t v_{\text{in}}(\tau) d\tau \quad (1.96)$$

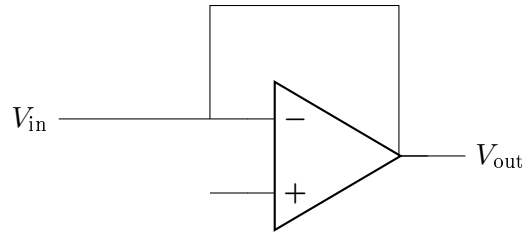


Figure 1.23: Op-amp integrator.

Differentiator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -RCs \quad (1.97)$$

Important Note

Pure differentiators amplify high-frequency noise. In practice, add a small resistor in series with C to limit high-frequency gain.

PID Controller Implementation

A PID controller can be implemented with op-amps:

$$G_{PID}(s) = K_p + \frac{K_i}{s} + K_d s \quad (1.98)$$

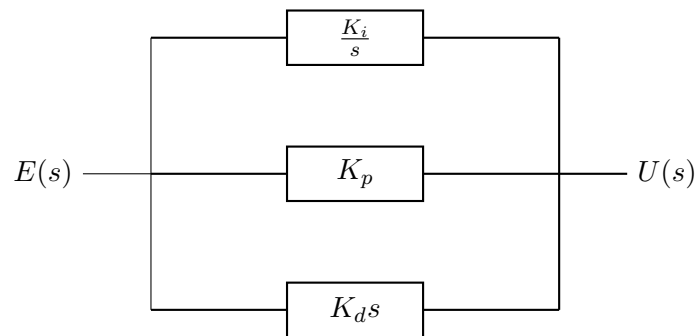


Figure 1.24: PID controller block diagram.

1.10.6 Active Filters

First-Order Low-Pass Filter

$$G(s) = \frac{K}{s/\omega_c + 1} \quad (1.99)$$

Passes low frequencies, attenuates high frequencies.

First-Order High-Pass Filter

$$G(s) = \frac{Ks/\omega_c}{s/\omega_c + 1} \quad (1.100)$$

Passes high frequencies, attenuates low frequencies (including DC).

Second-Order Butterworth Low-Pass

Maximally flat passband response:

$$G(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} \quad (1.101)$$

Damping ratio $\zeta = 1/\sqrt{2} \approx 0.707$, giving no overshoot in step response.

Band-Pass Filter

$$G(s) = \frac{K(\omega_0/Q)s}{s^2 + (\omega_0/Q)s + \omega_0^2} \quad (1.102)$$

Passes frequencies near ω_0 , attenuates both lower and higher frequencies.

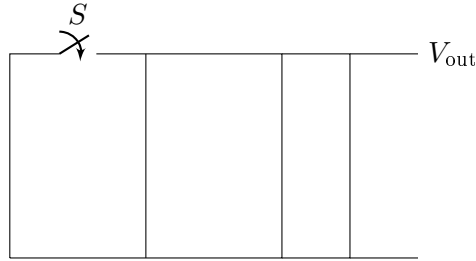
1.10.7 Power Electronics**Buck Converter (Step-Down)**

Figure 1.25: Buck converter circuit.

Steady-state output (ideal):

$$V_{\text{out}} = D \cdot V_{\text{in}} \quad (1.103)$$

where D is the duty cycle ($0 \leq D \leq 1$).

Small-signal transfer function (control to output):

$$\frac{\hat{v}_{\text{out}}(s)}{\hat{d}(s)} = V_{\text{in}} \cdot \frac{1}{LCs^2 + (L/R_L)s + 1} \quad (1.104)$$

1.10.8 Exercises: Electrical Systems

Exercise 1.6. For an RC circuit with $R = 47 \text{ k}\Omega$ and $C = 0.22 \text{ }\mu\text{F}$:

- (a) Find the time constant and cutoff frequency
- (b) Sketch the Bode plot (magnitude and phase)
- (c) Find the output voltage 5 ms after a 10V step input

Exercise 1.7. Design an RLC band-pass filter with center frequency 10 kHz and bandwidth 1 kHz. Calculate R , L , C , and Q .

Exercise 1.8. An op-amp integrator has $R = 100 \text{ k}\Omega$ and $C = 1 \text{ }\mu\text{F}$. If the input is $v_{\text{in}}(t) = 2 \sin(100t) \text{ V}$:

- (a) Find the transfer function
- (b) Calculate the steady-state output amplitude and phase
- (c) Write the time-domain output expression

Exercise 1.9. For the series RLC circuit, find the current $i(t)$ response to a unit step voltage input when $R = 100 \text{ }\Omega$, $L = 0.1 \text{ H}$, and $C = 100 \text{ }\mu\text{F}$. Is the circuit underdamped, critically damped, or overdamped?

Exercise 1.10. Design a second-order Sallen-Key low-pass filter with cutoff frequency 1 kHz and Butterworth response. Specify all component values.

1.11 Electromechanical Systems

Electromechanical systems convert between electrical and mechanical energy. They are ubiquitous in control applications—from servo motors in robots to voice coils in hard drives.

1.11.1 DC Motor Fundamentals

The DC motor is the workhorse of motion control, converting electrical power to rotational mechanical power.

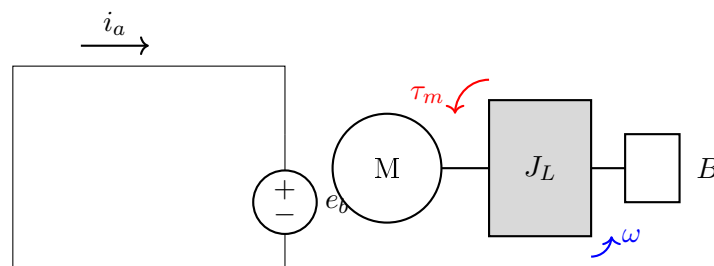


Figure 1.26: DC motor equivalent circuit and mechanical load.

Governing Equations

Electrical equation (armature circuit):

$$V_a = R_a i_a + L_a \frac{di_a}{dt} + e_b \quad (1.105)$$

Back-EMF (proportional to speed):

$$e_b = K_e \omega = K_e \dot{\theta} \quad (1.106)$$

Motor torque (proportional to current):

$$\tau_m = K_t i_a \quad (1.107)$$

Mechanical equation:

$$J\ddot{\theta} = \tau_m - B\dot{\theta} - \tau_L = K_t i_a - B\omega - \tau_L \quad (1.108)$$

where:

- V_a = armature voltage (V)
- R_a = armature resistance (Ω)
- L_a = armature inductance (H)
- i_a = armature current (A)
- e_b = back-EMF (V)
- K_e = back-EMF constant (V·s/rad)
- K_t = torque constant (N·m/A)
- J = total inertia (kg·m²)
- B = viscous friction (N·m·s/rad)
- $\omega = \dot{\theta}$ = angular velocity (rad/s)

Motor Constants

For an ideal DC motor: $K_t = K_e$ (in SI units). This follows from energy conservation—electrical power in equals mechanical power out.

Transfer Functions

Taking Laplace transforms:

$$V_a(s) = (R_a + L_a s)I_a(s) + K_e s\Theta(s) \quad (1.109)$$

$$Js^2\Theta(s) = K_t I_a(s) - Bs\Theta(s) - \tau_L(s) \quad (1.110)$$

Voltage to angular velocity (with $\tau_L = 0$):

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + B) + K_t K_e} \quad (1.111)$$

Voltage to angular position:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{s[(L_a s + R_a)(Js + B) + K_t K_e]} \quad (1.112)$$

Simplified Model (Neglecting Inductance)

For many motors, the electrical time constant $\tau_e = L_a/R_a$ is much smaller than the mechanical time constant $\tau_m = J/B$. Setting $L_a \approx 0$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t/R_a}{Js + B + K_tK_e/R_a} = \frac{K_m}{\tau_ms + 1} \quad (1.113)$$

where:

$$K_m = \frac{K_t}{R_aB + K_tK_e} = \frac{K_t}{K_tK_e + R_aB} \quad (\text{motor gain, rad/s/V}) \quad (1.114)$$

$$\tau_m = \frac{R_aJ}{R_aB + K_tK_e} \quad (\text{mechanical time constant}) \quad (1.115)$$

Example 1.8: DC Motor Analysis

A DC motor has the following parameters:

- $R_a = 2 \, \Omega$, $L_a = 0.5 \, \text{mH}$
- $K_t = K_e = 0.1 \, \text{N}\cdot\text{m/A} = 0.1 \, \text{V}\cdot\text{s/rad}$
- $J = 0.01 \, \text{kg}\cdot\text{m}^2$, $B = 0.001 \, \text{N}\cdot\text{m}\cdot\text{s/rad}$

Find the transfer function and steady-state speed for $V_a = 24 \, \text{V}$.

Solution:

Electrical time constant: $\tau_e = L_a/R_a = 0.0005/2 = 0.25 \, \text{ms}$

Since τ_e is small, use the simplified model:

$$K_m = \frac{0.1}{2 \times 0.001 + 0.1 \times 0.1} = \frac{0.1}{0.012} = 8.33 \, \text{rad/s/V} \quad (1.116)$$

$$\tau_m = \frac{2 \times 0.01}{0.012} = 1.67 \, \text{s} \quad (1.117)$$

Transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{8.33}{1.67s + 1} \quad (1.118)$$

Steady-state speed at 24V:

$$\omega_{ss} = K_m \times V_a = 8.33 \times 24 = 200 \, \text{rad/s} = 1910 \, \text{rpm} \quad (1.119)$$

1.11.2 DC Motor with Position Control

For position control applications, we need the voltage-to-position transfer function:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_m/s}{\tau_ms + 1} = \frac{K_m}{s(\tau_ms + 1)} \quad (1.120)$$

This is a Type 1 system (one free integrator), which can track step position commands with zero steady-state error.

State-Space Model

Choosing states $\mathbf{x} = [\theta, \omega, i_a]^\top$:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \\ \dot{i}_a \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B/J & K_t/J \\ 0 & -K_e/L_a & -R_a/L_a \end{bmatrix} \begin{bmatrix} \theta \\ \omega \\ i_a \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/L_a \end{bmatrix} V_a \quad (1.121)$$

Output (position): $y = \theta = [1, 0, 0]\mathbf{x}$

1.11.3 Brushless DC Motors (BLDC)

Brushless DC motors use electronic commutation instead of mechanical brushes. The control-relevant model is similar to the brushed DC motor:

$$\frac{\Omega(s)}{V_q(s)} = \frac{K_t/R}{(L/R)Js^2 + Js + K_tK_e/R} \quad (1.122)$$

where V_q is the q-axis voltage in the field-oriented control frame.

1.11.4 Stepper Motors

Stepper motors move in discrete angular steps, typically 1.8° (200 steps/revolution).

Open-loop position (idealized):

$$\theta = N_{\text{steps}} \times \theta_{\text{step}} \quad (1.123)$$

Dynamic model (simplified single phase):

$$J\ddot{\theta} + B\dot{\theta} + K_d \sin(N_r \theta) = K_t i \quad (1.124)$$

where N_r is the number of rotor teeth and K_d is the detent torque constant.

1.11.5 Voice Coil Actuators

Voice coils provide linear motion and are used in hard drives, loudspeakers, and precision positioning.

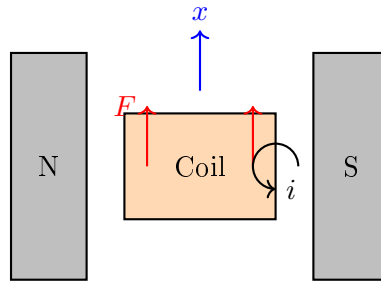


Figure 1.27: Voice coil actuator schematic.

Governing equations:

$$V = Ri + L \frac{di}{dt} + K_e \dot{x} \quad (1.125)$$

$$F = K_f i = m\ddot{x} + b\dot{x} \quad (1.126)$$

Transfer function (voltage to position):

$$\frac{X(s)}{V(s)} = \frac{K_f}{s[(Ls + R)(ms + b) + K_f K_e]} \quad (1.127)$$

1.11.6 Solenoids

Solenoids provide on/off linear actuation. The force depends nonlinearly on position:

$$F = \frac{1}{2} i^2 \frac{dL(x)}{dx} \quad (1.128)$$

where $L(x)$ is the position-dependent inductance.

Linearized about operating point (x_0, i_0) :

$$\Delta F = K_i \Delta i + K_x \Delta x \quad (1.129)$$

where K_i and K_x are partial derivatives of force with respect to current and position.

1.11.7 Piezoelectric Actuators

Piezoelectric actuators provide very small displacements (nm to μm) with high force and fast response.

Constitutive equations:

$$x = d_{33}V + s_{33}F/A \quad (1.130)$$

$$Q = CV + d_{33}F \quad (1.131)$$

where:

- d_{33} = piezoelectric coefficient (m/V or C/N)
- s_{33} = elastic compliance (m^2/N)
- C = capacitance (F)

Transfer function (simplified, free displacement):

$$\frac{X(s)}{V(s)} = \frac{d_{33}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (1.132)$$

1.11.8 Generators and Regeneration

When a motor is driven faster than its no-load speed, or when decelerating a load, it acts as a generator:

$$P_{\text{regen}} = e_b \cdot i_a = K_e \omega \cdot i_a \quad (1.133)$$

This regenerated power can be:

- Dissipated in braking resistors
- Returned to the power supply (regenerative braking)
- Stored in capacitors or batteries

1.11.9 Motor Selection Example

Example 1.9: Robot Joint Motor Selection

A robot joint must:

- Move a load with inertia $J_L = 0.05 \text{ kg}\cdot\text{m}^2$
- Achieve maximum velocity $\omega_{\max} = 10 \text{ rad/s}$
- Achieve maximum acceleration $\alpha_{\max} = 50 \text{ rad/s}^2$

Select an appropriate motor and gear ratio.

Solution:

Required torque at load:

$$\tau_L = J_L \alpha_{\max} = 0.05 \times 50 = 2.5 \text{ N}\cdot\text{m} \quad (1.134)$$

Consider a motor with:

- Rated torque: $\tau_m = 0.3 \text{ N}\cdot\text{m}$
- Rated speed: $3000 \text{ rpm} = 314 \text{ rad/s}$
- Rotor inertia: $J_m = 5 \times 10^{-5} \text{ kg}\cdot\text{m}^2$

Required gear ratio (from torque):

$$n \geq \frac{\tau_L}{\tau_m} = \frac{2.5}{0.3} = 8.3 \quad (1.135)$$

Check speed (with $n = 10$):

$$\omega_m = n \cdot \omega_L = 10 \times 10 = 100 \text{ rad/s} < 314 \text{ rad/s} \quad \checkmark \quad (1.136)$$

Reflected inertia check:

$$J_{\text{reflected}} = \frac{J_L}{n^2} = \frac{0.05}{100} = 5 \times 10^{-4} \text{ kg}\cdot\text{m}^2 \quad (1.137)$$

Total inertia at motor: $J_{\text{total}} = J_m + J_{\text{reflected}} = 5.5 \times 10^{-4} \text{ kg}\cdot\text{m}^2$

Motor acceleration:

$$\alpha_m = \frac{\tau_m}{J_{\text{total}}} = \frac{0.3}{5.5 \times 10^{-4}} = 545 \text{ rad/s}^2 \quad (1.138)$$

Load acceleration achieved:

$$\alpha_L = \frac{\alpha_m}{n} = \frac{545}{10} = 54.5 \text{ rad/s}^2 > 50 \text{ rad/s}^2 \quad \checkmark \quad (1.139)$$

The motor with 10:1 gear ratio meets all requirements.

1.11.10 Exercises: Electromechanical Systems

Exercise 1.11. A DC motor has $R_a = 1 \text{ } \Omega$, $K_t = K_e = 0.05 \text{ N}\cdot\text{m/A}$, $J = 2 \times 10^{-4} \text{ kg}\cdot\text{m}^2$, and $B = 10^{-4} \text{ N}\cdot\text{m}\cdot\text{s/rad}$. Neglect inductance.

- (a) Derive the transfer function from voltage to velocity

- (b) Find the steady-state speed for $V_a = 12$ V with no load
- (c) Find the stall torque (torque at zero speed) for $V_a = 12$ V

Exercise 1.12. For the motor in Exercise 1, a 5:1 gear reducer is added, and the load inertia is $J_L = 0.01$ kg·m² with load damping $B_L = 0.01$ N·m·s/rad.

- (a) Find the reflected load inertia and damping at the motor
- (b) Derive the new transfer function from voltage to load velocity
- (c) Compare the new time constant to the original

Exercise 1.13. A voice coil actuator has $R = 5$ Ω , $L = 1$ mH, $K_f = K_e = 10$ N/A, $m = 0.1$ kg, and $b = 2$ N·s/m.

- (a) Derive the transfer function from voltage to position
- (b) Find all poles and determine if the system is stable
- (c) Calculate the DC gain

Exercise 1.14. A piezoelectric actuator has $d_{33} = 500$ pm/V, resonant frequency 30 kHz, and damping ratio 0.02.

- (a) Write the transfer function
- (b) Find the static displacement for 100V input
- (c) Find the displacement amplitude at 20 kHz for 100V sinusoidal input

1.12 Thermal and Fluid Systems

Thermal and fluid systems are governed by energy and mass conservation. These systems often exhibit first-order dynamics with relatively slow time constants, making them excellent candidates for feedback control.

1.12.1 Thermal Systems Fundamentals

Heat Transfer Mechanisms

Conduction (through solids):

$$\dot{Q} = \frac{kA}{L}(T_1 - T_2) = \frac{T_1 - T_2}{R_{th}} \quad (1.140)$$

where $R_{th} = L/(kA)$ is the thermal resistance.

Convection (between solid and fluid):

$$\dot{Q} = hA(T_s - T_\infty) \quad (1.141)$$

where h is the convection coefficient (W/m²K).

Radiation:

$$\dot{Q} = \epsilon\sigma A(T_1^4 - T_2^4) \quad (1.142)$$

For small temperature differences, linearized: $\dot{Q} \approx h_r A(T_1 - T_2)$ where $h_r = 4\epsilon\sigma T_{avg}^3$.

Thermal Capacitance

The energy stored in a body:

$$E = mC_p T = \rho V C_p T \quad (1.143)$$

Rate of temperature change:

$$mC_p \frac{dT}{dt} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \quad (1.144)$$

1.12.2 Single-Capacitance Thermal System

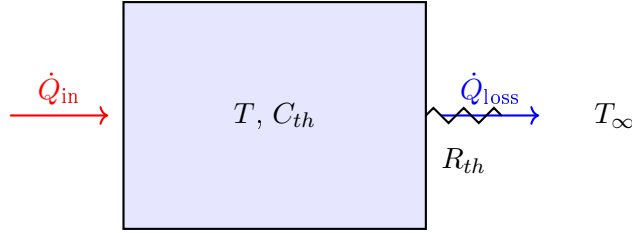


Figure 1.28: Single-capacitance thermal system.

Energy balance:

$$C_{th} \frac{dT}{dt} = \dot{Q}_{\text{in}} - \frac{T - T_{\infty}}{R_{th}} \quad (1.145)$$

where $C_{th} = mC_p$ is the thermal capacitance.

Rearranging:

$$\boxed{R_{th} C_{th} \frac{dT}{dt} + T = R_{th} \dot{Q}_{\text{in}} + T_{\infty}} \quad (1.146)$$

Transfer Function

Defining $\theta = T - T_{\infty}$ (temperature above ambient):

$$G(s) = \frac{\Theta(s)}{\dot{Q}_{\text{in}}(s)} = \frac{R_{th}}{\tau s + 1} \quad (1.147)$$

where $\tau = R_{th} C_{th}$ is the thermal time constant.

Example 1.10: Room Heating

A room has thermal capacitance $C_{th} = 500 \text{ kJ/K}$ and thermal resistance to outdoors $R_{th} = 0.01 \text{ K/W}$. Find:

- The thermal time constant
- The heater power needed to maintain 20°C when outdoor temperature is 0°C
- The room temperature 1 hour after turning off the heater

Solution:

(a) Time constant:

$$\tau = R_{th} C_{th} = 0.01 \times 500,000 = 5000 \text{ s} = 83.3 \text{ min} \quad (1.148)$$

(b) Steady-state heat balance:

$$\dot{Q}_{\text{in}} = \frac{T - T_{\infty}}{R_{th}} = \frac{20 - 0}{0.01} = 2000 \text{ W} = 2 \text{ kW} \quad (1.149)$$

(c) After heater off (cooling from 20°C):

$$T(t) = T_{\infty} + (T_0 - T_{\infty})e^{-t/\tau} = 0 + 20e^{-3600/5000} = 9.7^{\circ}\text{C} \quad (1.150)$$

1.12.3 Multi-Capacitance Thermal Systems

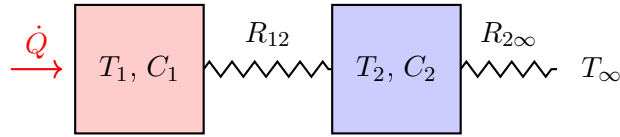


Figure 1.29: Two-capacitance thermal system.

Energy balances:

$$C_1 \frac{dT_1}{dt} = \dot{Q} - \frac{T_1 - T_2}{R_{12}} \quad (1.151)$$

$$C_2 \frac{dT_2}{dt} = \frac{T_1 - T_2}{R_{12}} - \frac{T_2 - T_{\infty}}{R_{2\infty}} \quad (1.152)$$

This yields a second-order system with two thermal time constants.

1.12.4 Fluid Systems: Tank Level

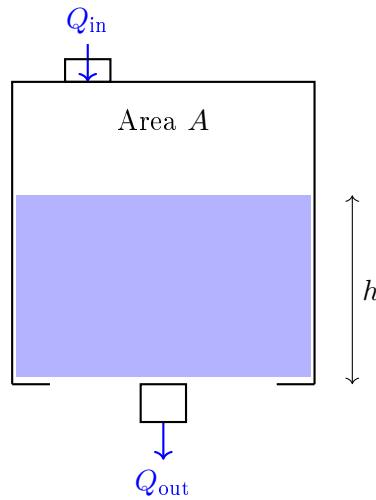


Figure 1.30: Single tank level system.

Mass Balance

For incompressible fluid:

$$A \frac{dh}{dt} = Q_{\text{in}} - Q_{\text{out}} \quad (1.153)$$

Outflow Models

Linear resistance (laminar flow through valve):

$$Q_{\text{out}} = \frac{h}{R_f} \quad (1.154)$$

where R_f is the fluid resistance.

Nonlinear resistance (turbulent flow, orifice):

$$Q_{\text{out}} = C_d A_o \sqrt{2gh} \quad (1.155)$$

This is nonlinear but can be linearized about an operating point.

Linear Tank Model

With linear resistance:

$$A \frac{dh}{dt} + \frac{h}{R_f} = Q_{\text{in}} \quad (1.156)$$

Transfer function:

$$G(s) = \frac{H(s)}{Q_{\text{in}}(s)} = \frac{R_f}{\tau s + 1} \quad (1.157)$$

where $\tau = AR_f$ is the hydraulic time constant.

Example 1.11: Water Tank

A cylindrical tank has diameter 2 m and drains through a valve with resistance $R_f = 100$ s/m². Find:

- (a) The time constant
- (b) Steady-state level for inflow of 0.01 m³/s
- (c) Time to reach 90% of steady-state from empty

Solution:

Tank area: $A = \pi(1)^2 = 3.14$ m²

(a) Time constant:

$$\tau = AR_f = 3.14 \times 100 = 314 \text{ s} = 5.2 \text{ min} \quad (1.158)$$

(b) Steady-state level:

$$h_{ss} = R_f \times Q_{\text{in}} = 100 \times 0.01 = 1 \text{ m} \quad (1.159)$$

(c) Time to 90%:

$$0.9 = 1 - e^{-t/\tau} \implies t = -\tau \ln(0.1) = 314 \times 2.3 = 723 \text{ s} = 12 \text{ min} \quad (1.160)$$

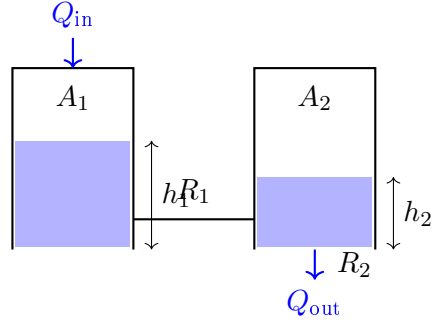


Figure 1.31: Two interacting tanks.

1.12.5 Interacting Tank Systems

Mass balances:

$$A_1 \frac{dh_1}{dt} = Q_{\text{in}} - \frac{h_1 - h_2}{R_1} \quad (1.161)$$

$$A_2 \frac{dh_2}{dt} = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2} \quad (1.162)$$

This is a second-order system. The interaction between tanks creates coupling that affects the system dynamics.

1.12.6 Hydraulic Actuators

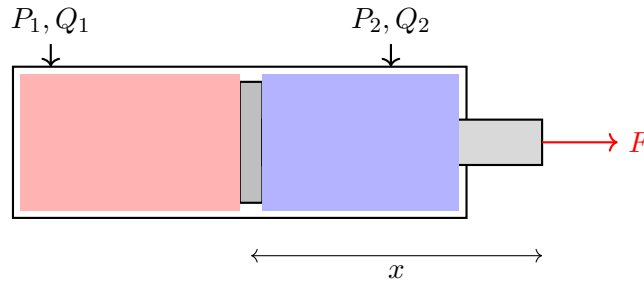


Figure 1.32: Double-acting hydraulic cylinder.

Flow Continuity

For incompressible fluid:

$$Q_1 = A_1 \dot{x} + \frac{V_1}{\beta} \dot{P}_1 \quad (1.163)$$

where β is the bulk modulus.

Force Balance

$$M\ddot{x} = P_1 A_1 - P_2 A_2 - F_{\text{load}} - B\dot{x} \quad (1.164)$$

Servo Valve Model

Flow through a servo valve:

$$Q = C_d w x_v \sqrt{\frac{2(P_s - P_L)}{\rho}} \quad (1.165)$$

where x_v is the valve spool position.

Linearized about operating point:

$$Q = K_q x_v - K_c P_L \quad (1.166)$$

1.12.7 Pneumatic Systems

Pneumatic systems use compressible gas (usually air), adding complexity:

Ideal gas law:

$$PV = mRT \quad (1.167)$$

Mass flow through orifice:

For subsonic flow ($P_2/P_1 > 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{2}{RT_1}} \sqrt{\frac{\gamma}{\gamma-1} \left[\left(\frac{P_2}{P_1} \right)^{2/\gamma} - \left(\frac{P_2}{P_1} \right)^{(\gamma+1)/\gamma} \right]} \quad (1.168)$$

For choked flow ($P_2/P_1 \leq 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{\gamma}{RT_1}} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(2(\gamma-1))} \quad (1.169)$$

Pneumatic Cylinder

Pressure dynamics in chamber:

$$\frac{dP}{dt} = \frac{\gamma RT}{V} (\dot{m}_{in} - \dot{m}_{out}) - \frac{\gamma P}{V} \dot{V} \quad (1.170)$$

1.12.8 Heat Exchangers

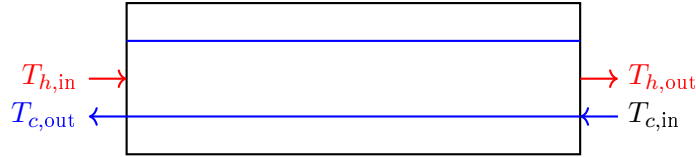


Figure 1.33: Counter-flow heat exchanger.

Simplified lumped model:

$$C_h \frac{dT_h}{dt} = \dot{m}_h c_{p,h} (T_{h,in} - T_h) - UA(T_h - T_c) \quad (1.171)$$

$$C_c \frac{dT_c}{dt} = \dot{m}_c c_{p,c} (T_{c,in} - T_c) + UA(T_h - T_c) \quad (1.172)$$

where UA is the overall heat transfer coefficient times area.

1.12.9 Exercises: Thermal and Fluid Systems

Exercise 1.15. A metal block ($m = 5$ kg, $C_p = 500$ J/kg·K) is heated by a 100 W heater and loses heat to 25°C ambient through convection ($hA = 5$ W/K).

- (a) Write the differential equation for block temperature
- (b) Find the thermal time constant
- (c) Find the steady-state temperature
- (d) How long to reach 90% of steady-state from ambient?

Exercise 1.16. Two tanks in series each have $A = 1$ m². The first tank drains into the second through resistance $R_1 = 50$ s/m², and the second drains to atmosphere through $R_2 = 100$ s/m².

- (a) Write the state-space model
- (b) Find the transfer function from Q_{in} to h_2
- (c) Find the system poles

Exercise 1.17. A hydraulic servo system has a 10 cm diameter piston, 1000 kg load mass, and bulk modulus $\beta = 1.5 \times 10^9$ Pa. The chamber volume is 0.5 L.

- (a) Calculate the hydraulic natural frequency
- (b) If damping ratio is 0.3, find the damping coefficient
- (c) Sketch the expected step response

Exercise 1.18. A hot water tank ($V = 200$ L) receives water at 15°C at rate 0.1 L/s and has a 3 kW heater. Heat loss to ambient is $UA = 10$ W/K. Find the steady-state temperature and time constant.

1.13 Chemical Systems

Chemical processes involve mass and energy balances coupled with reaction kinetics. These systems often exhibit nonlinear behavior, time delays, and complex interactions, making them challenging but important control applications.

1.13.1 Continuous Stirred-Tank Reactor (CSTR)

The CSTR is a fundamental unit in chemical engineering and a classic control problem.

Assumptions

- Perfect mixing (uniform concentration and temperature)
- Constant volume V and density ρ
- Constant flow rate F (in = out)
- Single first-order irreversible reaction: $A \rightarrow B$

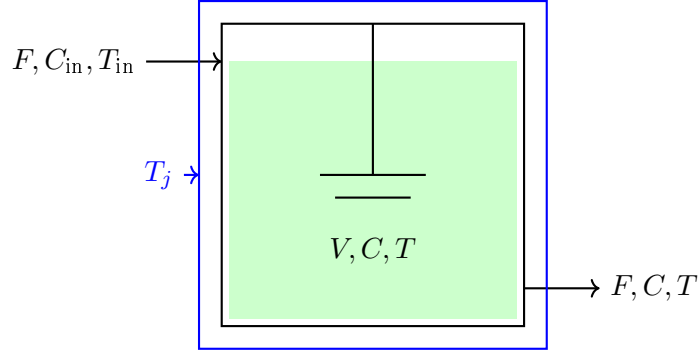


Figure 1.34: Continuous stirred-tank reactor with cooling jacket.

Component Mass Balance

$$V \frac{dC_A}{dt} = F(C_{A,\text{in}} - C_A) - V k(T) C_A \quad (1.173)$$

where $k(T)$ is the temperature-dependent reaction rate constant.

Arrhenius Equation

$$k(T) = k_0 \exp\left(-\frac{E_a}{RT}\right) \quad (1.174)$$

where:

- k_0 = pre-exponential factor (1/s)
- E_a = activation energy (J/mol)
- R = universal gas constant (8.314 J/mol·K)

Energy Balance

$$\rho V C_p \frac{dT}{dt} = F \rho C_p (T_{\text{in}} - T) + (-\Delta H_r) V k(T) C_A - U A (T - T_j) \quad (1.175)$$

where:

- ΔH_r = heat of reaction (J/mol, negative for exothermic)
- $U A$ = heat transfer coefficient times area (W/K)
- T_j = jacket temperature (K)

Linearized Model

Linearizing about steady-state (C_{A0}, T_0):

$$\frac{d\Delta C_A}{dt} = a_{11} \Delta C_A + a_{12} \Delta T + b_1 \Delta C_{A,\text{in}} \quad (1.176)$$

$$\frac{d\Delta T}{dt} = a_{21} \Delta C_A + a_{22} \Delta T + b_2 \Delta T_j \quad (1.177)$$

State-space form:

$$\begin{bmatrix} \Delta \dot{C}_A \\ \Delta \dot{T} \end{bmatrix} = \mathbf{A} \begin{bmatrix} \Delta C_A \\ \Delta T \end{bmatrix} + \mathbf{B} \begin{bmatrix} \Delta C_{A,\text{in}} \\ \Delta T_j \end{bmatrix} \quad (1.178)$$

Example 1.12: CSTR Linearization

A CSTR operates at steady-state with:

- $V = 1 \text{ m}^3$, $F = 0.1 \text{ m}^3/\text{min}$
- $C_{A0} = 2 \text{ mol/L}$, $T_0 = 350 \text{ K}$
- $k_0 = 10^{10} \text{ min}^{-1}$, $E_a/R = 8750 \text{ K}$

The linearized $\mathbf{A}$ matrix elements are:

$$a_{11} = -\frac{F}{V} - k(T_0) = -0.1 - k_0 e^{-8750/350} = -0.1 - 1.2 = -1.3 \text{ min}^{-1} \quad (1.179)$$

$$a_{12} = -C_{A0} \left. \frac{dk}{dT} \right|_{T_0} = -C_{A0} k(T_0) \frac{E_a}{RT_0^2} = -0.17 \text{ mol/L} \cdot \text{K} \cdot \text{min} \quad (1.180)$$

The negative a_{12} indicates that increased temperature increases reaction rate, consuming more C_A .

1.13.2 Mixing Tank

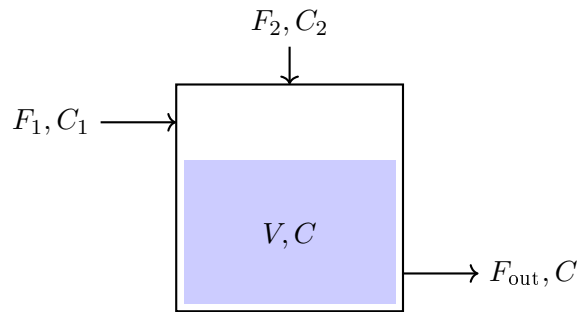


Figure 1.35: Mixing tank with two inlets.

Mass Balance (Total)

$$\frac{d(\rho V)}{dt} = \rho_1 F_1 + \rho_2 F_2 - \rho F_{\text{out}} \quad (1.181)$$

For constant density and volume: $F_{\text{out}} = F_1 + F_2$

Component Balance

$$V \frac{dC}{dt} = F_1 C_1 + F_2 C_2 - (F_1 + F_2) C \quad (1.182)$$

Transfer function (for constant F_1, F_2):

$$\frac{C(s)}{C_1(s)} = \frac{F_1/V}{s + (F_1 + F_2)/V} = \frac{F_1/(F_1 + F_2)}{\tau s + 1} \quad (1.183)$$

where $\tau = V/(F_1 + F_2)$ is the residence time.

1.13.3 pH Neutralization

pH control is notoriously difficult due to the highly nonlinear relationship between acid/base concentration and pH.

$$\text{pH} = -\log_{10}[H^+] \quad (1.184)$$

For a strong acid-strong base system:

$$[H^+] = \frac{C_a - C_b + \sqrt{(C_a - C_b)^2 + 4K_w}}{2} \quad (1.185)$$

where $K_w = 10^{-14}$ at 25°C.

Titration curve: The pH changes dramatically near the equivalence point, creating a high-gain nonlinearity.

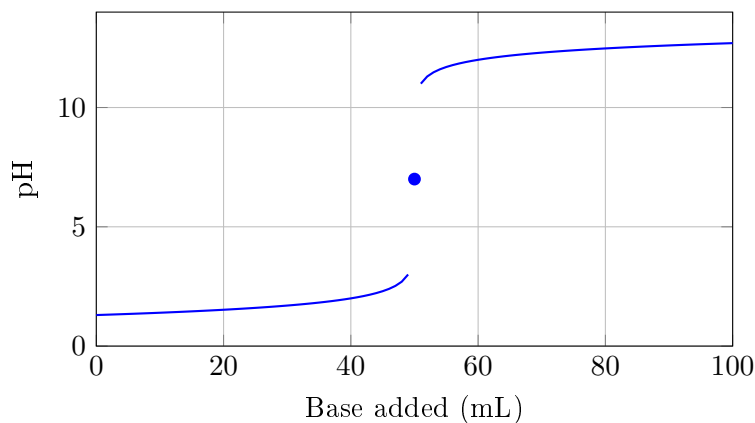


Figure 1.36: Typical pH titration curve showing high gain near equivalence.

1.13.4 Distillation Column

Distillation separates mixtures based on volatility differences.

Simplified Model

For a binary mixture, the composition dynamics on tray n :

$$M_n \frac{dx_n}{dt} = L_{n-1}x_{n-1} + V_{n+1}y_{n+1} - L_nx_n - V_ny_n \quad (1.186)$$

where:

- M_n = liquid holdup on tray

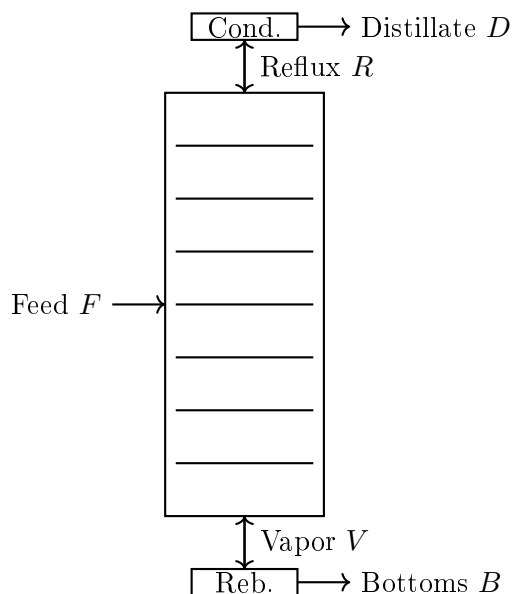


Figure 1.37: Distillation column schematic.

- L_n = liquid flow rate
- V_n = vapor flow rate
- x_n = liquid composition (mole fraction)
- y_n = vapor composition

Vapor-liquid equilibrium:

$$y_n = \frac{\alpha x_n}{1 + (\alpha - 1)x_n} \quad (1.187)$$

where α is the relative volatility.

1.13.5 Batch Reactor

Unlike continuous reactors, batch reactors have no flow during reaction:

$$\frac{dC_A}{dt} = -k(T)C_A \quad (1.188)$$

$$\frac{dT}{dt} = \frac{(-\Delta H_r)k(T)C_A}{\rho C_p} - \frac{UA(T - T_j)}{\rho V C_p} \quad (1.189)$$

The concentration decreases exponentially (for first-order kinetics):

$$C_A(t) = C_{A0} \exp \left(- \int_0^t k(T(\tau)) d\tau \right) \quad (1.190)$$

1.13.6 Reaction Kinetics

Zero-Order Reaction

$$\frac{dC}{dt} = -k \quad (1.191)$$

Solution: $C(t) = C_0 - kt$

First-Order Reaction

$$\frac{dC}{dt} = -kC \quad (1.192)$$

Solution: $C(t) = C_0 e^{-kt}$

Half-life: $t_{1/2} = \ln(2)/k$

Second-Order Reaction

$$\frac{dC}{dt} = -kC^2 \quad (1.193)$$

Solution: $\frac{1}{C(t)} = \frac{1}{C_0} + kt$

Enzyme Kinetics (Michaelis-Menten)

$$\frac{dC}{dt} = -\frac{V_{\max}C}{K_m + C} \quad (1.194)$$

where $V_{\max}$ is maximum rate and K_m is the Michaelis constant.

1.13.7 Process Dead Time

Many chemical processes exhibit significant time delays (dead time, transport lag):

$$y(t) = u(t - \theta) \quad (1.195)$$

Transfer function:

$$G(s) = e^{-\theta s} \quad (1.196)$$

Dead time makes control more challenging. Common approximations:

First-order Padé:

$$e^{-\theta s} \approx \frac{1 - \theta s/2}{1 + \theta s/2} \quad (1.197)$$

Second-order Padé:

$$e^{-\theta s} \approx \frac{1 - \theta s/2 + (\theta s)^2/12}{1 + \theta s/2 + (\theta s)^2/12} \quad (1.198)$$

Example 1.13: First-Order Plus Dead Time (FOPDT)

Many chemical processes can be approximated as:

$$G(s) = \frac{K e^{-\theta s}}{\tau s + 1} \quad (1.199)$$

For a heat exchanger with $K = 2\text{ }^\circ\text{C}/\%$, $\tau = 60\text{ s}$, $\theta = 10\text{ s}$:

$$G(s) = \frac{2e^{-10s}}{60s + 1} \quad (1.200)$$

The dead-time-to-time-constant ratio $\theta/\tau = 10/60 = 0.17$ indicates this system is relatively easy to control.

1.13.8 Exercises: Chemical Systems

Exercise 1.19. A CSTR with volume 500 L and flow rate 50 L/min processes a first-order reaction with $k = 0.1\text{ min}^{-1}$. Feed concentration is 2 mol/L.

- (a) Find the steady-state exit concentration
- (b) Find the residence time
- (c) Write the transfer function from C_{in} to C
- (d) Find the time constant

Exercise 1.20. A mixing tank blends two streams: Stream 1 at 100 L/min with 10% salt, and Stream 2 at 50 L/min with 2% salt. Tank volume is 1000 L.

- (a) Find the steady-state exit concentration
- (b) If Stream 1 concentration suddenly changes to 15%, find the new steady-state and the time to reach 90% of the change

Exercise 1.21. An exothermic batch reactor starts at 25°C with 1000 mol of reactant. Heat of reaction is -50 kJ/mol , heat capacity is $4\text{ kJ/kg}\cdot\text{K}$, mass is 1000 kg, and reaction rate constant is $k = 0.01\text{ min}^{-1}$ at 25°C with $E_a/R = 5000\text{ K}$.

- (a) Write the coupled ODEs for concentration and temperature (adiabatic case)
- (b) Estimate the adiabatic temperature rise if reaction goes to completion

Exercise 1.22. A process has FOPDT model with $K = 1.5$, $\tau = 120\text{ s}$, and $\theta = 30\text{ s}$.

- (a) Write the transfer function
- (b) Calculate θ/τ and comment on controllability
- (c) Use first-order Padé approximation and find the system poles

1.14 Aerospace Systems

Aerospace systems present unique control challenges: operation in three dimensions, coupling between axes, varying dynamics with flight conditions, and stringent safety requirements.

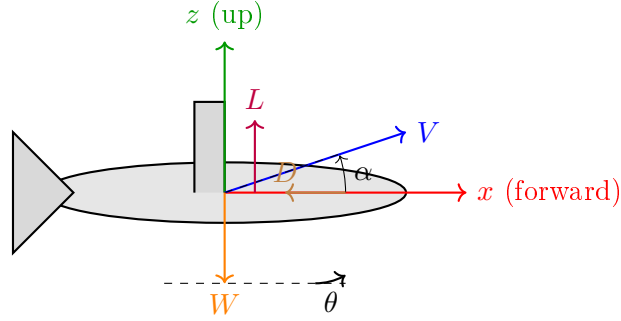


Figure 1.38: Aircraft longitudinal plane showing forces and angles.

1.14.1 Aircraft Longitudinal Dynamics

State Variables

- u = forward velocity perturbation (m/s)
- w = vertical velocity perturbation (m/s)
- q = pitch rate (rad/s)
- θ = pitch angle (rad)

Linearized Equations of Motion

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & U_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} \\ 0 \end{bmatrix} \delta_e \quad (1.201)$$

where δ_e is elevator deflection and X_u , Z_w , etc. are stability derivatives.

Characteristic Modes

The longitudinal dynamics typically have two modes:

Short Period Mode:

- Fast oscillation (1-3 seconds period)
- Primarily α and q motion
- Well-damped ($\zeta \approx 0.3 - 0.7$)

Approximate transfer function:

$$\frac{\alpha(s)}{\delta_e(s)} \approx \frac{Z_{\delta_e}/U_0}{s^2 - (M_q + M_{\dot{\alpha}} + Z_w/U_0)s + (Z_w M_q/U_0 - M_w)} \quad (1.202)$$

Phugoid Mode:

- Slow oscillation (30-100 seconds period)
- Exchange between kinetic and potential energy

- Lightly damped ($\zeta \approx 0.04 - 0.1$)

Approximate frequency and damping:

$$\omega_p \approx \sqrt{2} \frac{g}{U_0} \quad (1.203)$$

$$\zeta_p \approx \frac{1}{\sqrt{2}} \frac{C_D}{C_L} \quad (1.204)$$

Example 1.14: Boeing 747 Longitudinal Modes

At cruise ($U_0 = 250$ m/s, altitude 10 km):

Short period:

- $\omega_{sp} = 0.9$ rad/s, $\zeta_{sp} = 0.35$
- Period: $T_{sp} = 2\pi/\omega_{sp} = 7$ s

Phugoid:

- $\omega_p = 0.06$ rad/s, $\zeta_p = 0.05$
- Period: $T_p = 2\pi/\omega_p = 105$ s

1.14.2 Aircraft Lateral-Directional Dynamics

State Variables

- β = sideslip angle (rad)
- p = roll rate (rad/s)
- r = yaw rate (rad/s)
- ϕ = bank angle (rad)

Characteristic Modes

Dutch Roll:

- Coupled yaw-roll oscillation
- Period: 3-10 seconds
- Damping varies with aircraft design

Roll Mode:

- Pure rolling motion
- First-order, typically fast (time constant 0.5-2 s)

Spiral Mode:

- Slow divergence or convergence
- Often slightly unstable (acceptable if slow)

1.14.3 Rocket Dynamics

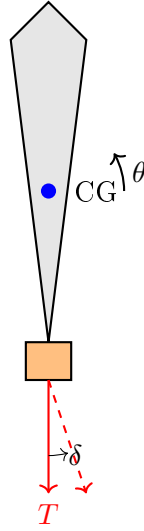


Figure 1.39: Rocket with thrust vector control.

Pitch Dynamics (Simplified)

For a rocket with thrust vector control:

$$I_{yy}\ddot{\theta} = T \cdot \ell_T \cdot \delta \quad (1.205)$$

where:

- I_{yy} = pitch moment of inertia
- T = thrust
- ℓ_T = distance from CG to gimbal point
- δ = gimbal angle

Transfer function:

$$\frac{\Theta(s)}{\delta(s)} = \frac{T\ell_T/I_{yy}}{s^2} = \frac{K_r}{s^2} \quad (1.206)$$

This is a double integrator—marginally stable and requires feedback for control.

Aerodynamic Effects

With aerodynamic forces, additional terms appear:

$$I_{yy}\ddot{\theta} = T\ell_T\delta + M_\alpha\alpha + M_qq \quad (1.207)$$

For a statically unstable rocket ($M_\alpha > 0$), the system has a pole in the right half-plane.

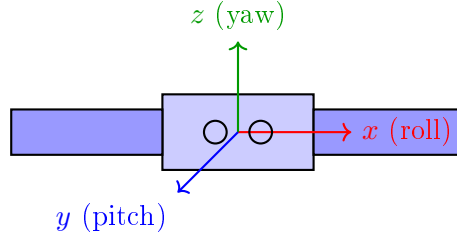


Figure 1.40: Satellite with body-fixed axes.

1.14.4 Satellite Attitude Dynamics

Euler's Equations

$$I_{xx}\dot{\omega}_x + (I_{zz} - I_{yy})\omega_y\omega_z = \tau_x \quad (1.208)$$

$$I_{yy}\dot{\omega}_y + (I_{xx} - I_{zz})\omega_z\omega_x = \tau_y \quad (1.209)$$

$$I_{zz}\dot{\omega}_z + (I_{yy} - I_{xx})\omega_x\omega_y = \tau_z \quad (1.210)$$

Linearized Single-Axis Model

For small angles about one axis:

$$I\ddot{\theta} = \tau \quad (1.211)$$

Transfer function:

$$\frac{\Theta(s)}{\tau(s)} = \frac{1}{Is^2} \quad (1.212)$$

Reaction Wheel Dynamics

A reaction wheel applies torque by changing its angular momentum:

$$\tau_{\text{satellite}} = -I_w\dot{\omega}_w \quad (1.213)$$

Combined system:

$$I_s\ddot{\theta} = -I_w\dot{\omega}_w = I_w\ddot{\theta}_w \quad (1.214)$$

1.14.5 Quadrotor Dynamics

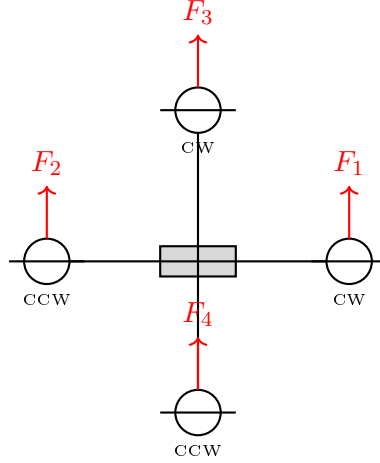
Control Inputs

$$\text{Total thrust: } T = F_1 + F_2 + F_3 + F_4 \quad (1.215)$$

$$\text{Roll torque: } \tau_\phi = \ell(F_1 - F_2) \quad (1.216)$$

$$\text{Pitch torque: } \tau_\theta = \ell(F_3 - F_4) \quad (1.217)$$

$$\text{Yaw torque: } \tau_\psi = k_\tau(F_1 + F_2 - F_3 - F_4) \quad (1.218)$$



Roll: $F_1 \neq F_2$, Pitch: $F_3 \neq F_4$, Yaw: $\text{CW} \neq \text{CCW}$

Figure 1.41: Quadrotor configuration.

Equations of Motion

Translational (in body frame):

$$m\ddot{x} = -mg \sin \theta \quad (1.219)$$

$$m\ddot{y} = mg \cos \theta \sin \phi \quad (1.220)$$

$$m\ddot{z} = mg \cos \theta \cos \phi - T \quad (1.221)$$

Rotational:

$$I_{xx}\ddot{\phi} = \tau_{\phi} \quad (1.222)$$

$$I_{yy}\ddot{\theta} = \tau_{\theta} \quad (1.223)$$

$$I_{zz}\ddot{\psi} = \tau_{\psi} \quad (1.224)$$

Linearized Model (Hover)

Near hover ($\phi, \theta, \psi \approx 0$):

$$\ddot{\phi} = \frac{\ell}{I_{xx}}(F_1 - F_2), \quad \ddot{\theta} = \frac{\ell}{I_{yy}}(F_3 - F_4), \quad \ddot{z} = g - \frac{T}{m} \quad (1.225)$$

1.14.6 Spacecraft Orbital Mechanics

Hill-Clohessy-Wiltshire Equations

Relative motion near a circular reference orbit:

$$\ddot{x} - 2n\dot{y} - 3n^2x = a_x \quad (1.226)$$

$$\ddot{y} + 2n\dot{x} = a_y \quad (1.227)$$

$$\ddot{z} + n^2z = a_z \quad (1.228)$$

where $n = \sqrt{\mu/a^3}$ is the orbital rate.

The out-of-plane motion (z) is decoupled and is a simple harmonic oscillator.
 The in-plane motion (x, y) is coupled, with the solution:

$$x(t) = (4 - 3 \cos nt)x_0 + \frac{\sin nt}{n}\dot{x}_0 + \frac{2(1 - \cos nt)}{n}\dot{y}_0 \quad (1.229)$$

1.14.7 Exercises: Aerospace Systems

Exercise 1.23. An aircraft has short-period approximation:

$$\frac{\alpha(s)}{\delta_e(s)} = \frac{-2}{s^2 + 1.5s + 4} \quad (1.230)$$

- (a) Find the natural frequency and damping ratio
- (b) Is this response acceptable for a fighter aircraft?
- (c) Sketch the step response

Exercise 1.24. A rocket has pitch moment of inertia $I_{yy} = 10^6 \text{ kg}\cdot\text{m}^2$, thrust $T = 10^6 \text{ N}$, and gimbal arm $\ell_T = 20 \text{ m}$.

- (a) Find the transfer function from gimbal angle to pitch rate
- (b) Design a proportional controller to achieve 1 rad/s^2 pitch acceleration per degree of gimbal

Exercise 1.25. A satellite reaction wheel has inertia $I_w = 0.01 \text{ kg}\cdot\text{m}^2$ and the satellite has inertia $I_s = 100 \text{ kg}\cdot\text{m}^2$ about one axis.

- (a) If the wheel accelerates at 100 rad/s^2 , find the satellite angular acceleration
- (b) Find the maximum satellite rate if the wheel is limited to 6000 rpm

Exercise 1.26. A quadrotor has mass 1.5 kg, arm length 0.25 m, and moment of inertia $I_{xx} = I_{yy} = 0.02 \text{ kg}\cdot\text{m}^2$.

- (a) Find the hover thrust per motor
- (b) Derive the transfer function from motor force differential to roll angle
- (c) If motor response is modeled as first-order with $\tau = 0.05 \text{ s}$, find the complete roll transfer function

1.15 Biomedical Systems

Biomedical control systems interface with the human body, requiring special consideration for safety, variability between patients, and the inherent complexity of biological processes.

1.15.1 Pharmacokinetics: Drug Distribution

Pharmacokinetics describes how drugs move through the body.

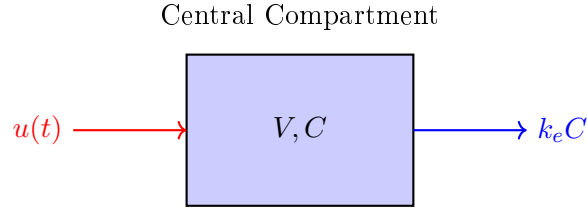


Figure 1.42: One-compartment pharmacokinetic model.

One-Compartment Model

Mass balance:

$$V \frac{dC}{dt} = u(t) - k_e V C \quad (1.231)$$

where:

- V = volume of distribution (L)
- C = drug concentration (mg/L)
- $u(t)$ = infusion rate (mg/min)
- k_e = elimination rate constant (1/min)

Transfer function:

$$\frac{C(s)}{U(s)} = \frac{1/V}{s + k_e} \quad (1.232)$$

Half-life: $t_{1/2} = \ln(2)/k_e$

Example 1.15: IV Drug Infusion

A drug has $V = 50$ L and $k_e = 0.1 \text{ min}^{-1}$. Find:

- (a) The half-life
- (b) The infusion rate needed for steady-state concentration of 10 mg/L
- (c) Time to reach 90% of steady-state

Solution:

- (a) Half-life: $t_{1/2} = \ln(2)/0.1 = 6.93 \text{ min}$
- (b) At steady-state: $u_{ss} = k_e V C_{ss} = 0.1 \times 50 \times 10 = 50 \text{ mg/min}$
- (c) Time constant: $\tau = 1/k_e = 10 \text{ min}$
Time to 90%: $t = -\tau \ln(0.1) = 10 \times 2.3 = 23 \text{ min}$

Two-Compartment Model

Mass balances:

$$V_1 \frac{dC_1}{dt} = u(t) - (k_{10} + k_{12})V_1 C_1 + k_{21}V_2 C_2 \quad (1.233)$$

$$V_2 \frac{dC_2}{dt} = k_{12}V_1 C_1 - k_{21}V_2 C_2 \quad (1.234)$$

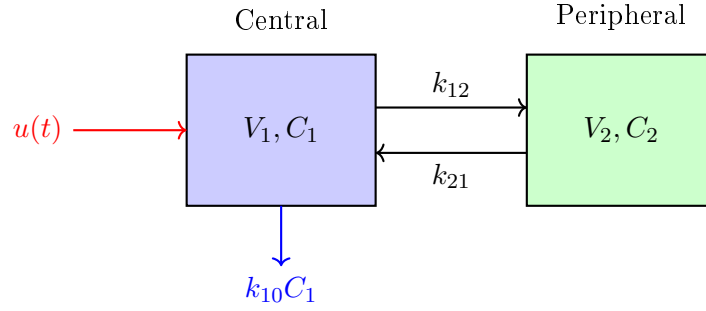


Figure 1.43: Two-compartment pharmacokinetic model.

This yields a second-order system with two exponential phases:

- **Distribution phase** (α): rapid initial decline
- **Elimination phase** (β): slower terminal decline

1.15.2 Glucose-Insulin Dynamics

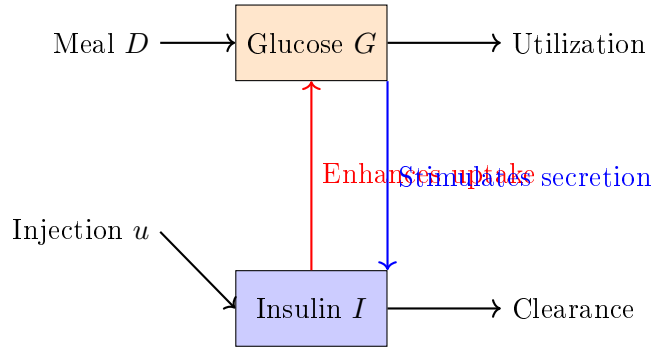


Figure 1.44: Glucose-insulin regulatory system.

Minimal Model (Bergman)

$$\frac{dG}{dt} = -p_1 G - X(G + G_b) + D(t) \quad (1.235)$$

$$\frac{dX}{dt} = -p_2 X + p_3(I - I_b) \quad (1.236)$$

$$\frac{dI}{dt} = -n(I - I_b) + \gamma(G - h)^+ + u(t) \quad (1.237)$$

where:

- G = glucose concentration (mg/dL)
- I = insulin concentration (μ U/mL)
- X = insulin action (remote compartment)

- $D(t)$ = meal disturbance
- $u(t)$ = insulin infusion (for artificial pancreas)

Linearized Model for Control Design

Near operating point (G_0, I_0) :

$$\begin{bmatrix} \Delta \dot{G} \\ \Delta \dot{I} \end{bmatrix} = \begin{bmatrix} -p_1 - X_0 & -K_G \\ K_I & -n \end{bmatrix} \begin{bmatrix} \Delta G \\ \Delta I \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Delta D + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta u \quad (1.238)$$

1.15.3 Cardiovascular System

Windkessel Model

The arterial system can be modeled as an electrical analog:

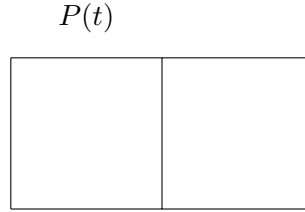


Figure 1.45: Two-element Windkessel model.

Governing equation:

$$C \frac{dP}{dt} + \frac{P}{R} = Q(t) \quad (1.239)$$

where:

- P = arterial pressure (mmHg)
- Q = cardiac output (mL/s)
- C = arterial compliance (mL/mmHg)
- R = systemic vascular resistance (mmHg·s/mL)

Blood Pressure Control

The baroreflex regulates blood pressure:

$$\frac{dP}{dt} = -\frac{1}{\tau_P} (P - P_{\text{set}}) + K_u u(t) \quad (1.240)$$

where $u(t)$ represents vasoactive drug infusion.

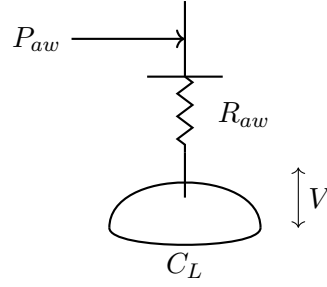


Figure 1.46: Single-compartment lung model.

1.15.4 Respiratory System

Mechanical Ventilation Model

Equation of motion:

$$P_{aw} = R_{aw}\dot{V} + \frac{V}{C_L} + P_{PEEP} \quad (1.241)$$

where:

- P_{aw} = airway pressure (cmH₂O)
- R_{aw} = airway resistance (cmH₂O·s/L)
- C_L = lung compliance (L/cmH₂O)
- V = lung volume (L)
- P_{PEEP} = positive end-expiratory pressure

Transfer function:

$$\frac{V(s)}{P_{aw}(s)} = \frac{C_L}{R_{aw}C_L s + 1} \quad (1.242)$$

1.15.5 Anesthesia Delivery

Propofol Pharmacokinetics

Three-compartment mammillary model:

$$V_1\dot{C}_1 = u - k_{10}V_1C_1 - k_{12}V_1C_1 + k_{21}V_2C_2 - k_{13}V_1C_1 + k_{31}V_3C_3 \quad (1.243)$$

$$V_2\dot{C}_2 = k_{12}V_1C_1 - k_{21}V_2C_2 \quad (1.244)$$

$$V_3\dot{C}_3 = k_{13}V_1C_1 - k_{31}V_3C_3 \quad (1.245)$$

Pharmacodynamic Effect

The effect-site concentration C_e relates to the measured output (e.g., BIS index):

$$\frac{dC_e}{dt} = k_{e0}(C_1 - C_e) \quad (1.246)$$

Hill equation for effect:

$$\text{Effect} = E_0 - E_{\max} \frac{C_e^\gamma}{C_e^\gamma + EC_{50}^\gamma} \quad (1.247)$$

1.15.6 Medical Device Examples

Infusion Pump

Transfer function (actuator dynamics):

$$\frac{Q(s)}{V_{\text{cmd}}(s)} = \frac{K_p}{\tau_p s + 1} \quad (1.248)$$

where Q is flow rate and V_{cmd} is the command voltage.

Pacemaker Rate Response

Activity-based pacing rate:

$$\tau \frac{dR}{dt} + R = R_{\text{base}} + K \cdot A(t) \quad (1.249)$$

where $A(t)$ is the activity sensor signal and R is the pacing rate.

Cochlear Implant Signal Processing

Filter bank model:

$$H_k(s) = \frac{\omega_k^2}{s^2 + 2\zeta\omega_k s + \omega_k^2} \quad (1.250)$$

for frequency band k centered at ω_k .

1.15.7 Neural Systems

Hodgkin-Huxley Neuron Model

$$C_m \frac{dV}{dt} = I_{\text{ext}} - g_{Na} m^3 h (V - E_{Na}) - g_K n^4 (V - E_K) - g_L (V - E_L) \quad (1.251)$$

with gating variables m , h , n following first-order kinetics.

Simplified Integrate-and-Fire Model

$$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + R_m I_{\text{ext}} \quad (1.252)$$

When V reaches threshold V_{th} , the neuron fires and resets.

1.15.8 Exercises: Biomedical Systems

Exercise 1.27. A drug has one-compartment kinetics with $V = 35$ L and $t_{1/2} = 4$ hours.

- Calculate the elimination rate constant
- Find the loading dose to achieve 20 mg/L immediately
- Find the maintenance infusion rate
- Write the transfer function from infusion rate to concentration

Exercise 1.28. A patient's arterial compliance is $C = 2$ mL/mmHg and peripheral resistance is $R = 1$ mmHg·s/mL.

- (a) Find the time constant of the arterial system
- (b) If cardiac output steps from 80 to 100 mL/s, find the new steady-state pressure
- (c) Sketch the pressure response to the step change

Exercise 1.29. A mechanical ventilator delivers pressure to a patient with $R_{aw} = 10 \text{ cmH}_2\text{O}\cdot\text{s/L}$ and $C_L = 0.05 \text{ L/cmH}_2\text{O}$.

- (a) Find the time constant
- (b) If inspiratory pressure is 20 cmH₂O above PEEP, find the tidal volume
- (c) Find the inspiratory time needed to achieve 95% of the tidal volume

Exercise 1.30. An artificial pancreas controller must regulate glucose. The simplified model has:

$$\frac{G(s)}{U(s)} = \frac{-2}{(10s + 1)(50s + 1)} \quad (1.253)$$

where G is glucose (mg/dL) and U is insulin (U/hr), with time in minutes.

- (a) Find the steady-state glucose change for 1 U/hr insulin increase
- (b) Find the system poles and time constants
- (c) Sketch the glucose response to a step insulin input

1.16 Industrial and Computer Systems

This section covers systems from manufacturing, process control, and computer science—domains where control theory meets discrete events, queueing, and information flow.

1.16.1 Industrial Process Control

Conveyor Belt System

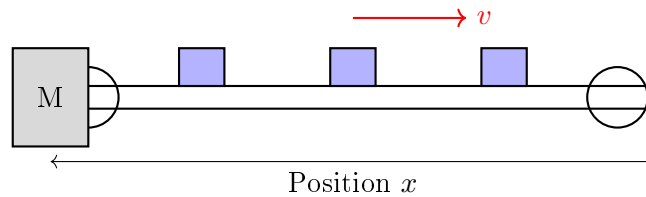


Figure 1.47: Conveyor belt system with motor drive.

Motor-driven conveyor dynamics:

$$J_{eq}\dot{\omega} + B\omega = K_t i - \tau_L \quad (1.254)$$

where J_{eq} includes reflected inertia of belt and load.

Belt velocity: $v = r\omega$ where r is the drive roller radius.

Transfer function (speed control):

$$\frac{V(s)}{I(s)} = \frac{rK_t}{J_{eq}s + B} \quad (1.255)$$

Temperature Control Zone

Industrial ovens often have multiple zones:

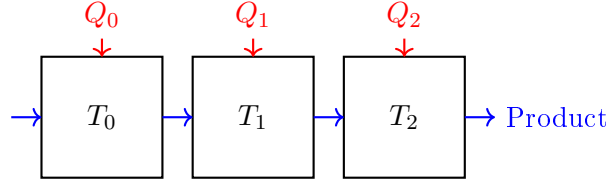


Figure 1.48: Multi-zone temperature control.

Each zone:

$$C_i \frac{dT_i}{dt} = Q_i - \frac{T_i - T_{i-1}}{R_{i-1,i}} - \frac{T_i - T_{i+1}}{R_{i,i+1}} - \frac{T_i - T_\infty}{R_{i,\infty}} \quad (1.256)$$

1.16.2 Robotic Manipulators

Single-Link Robot Arm

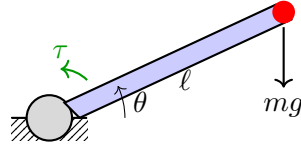


Figure 1.49: Single-link robot arm.

Equation of motion:

$$J\ddot{\theta} + B\dot{\theta} + mgl \cos \theta = \tau \quad (1.257)$$

Linearized about horizontal ($\theta_0 = 0$):

$$J\ddot{\theta} + B\dot{\theta} + mgl = \tau \quad (1.258)$$

Note: Gravity creates a constant disturbance torque.

Two-Link Planar Arm

The dynamics become coupled and nonlinear:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} \quad (1.259)$$

where:

- $\mathbf{M}(\mathbf{q})$ = mass matrix (configuration-dependent)
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ = Coriolis and centrifugal terms
- $\mathbf{g}(\mathbf{q})$ = gravity vector
- $\boldsymbol{\tau}$ = joint torques

For a two-link arm:

$$M_{11} = m_1 \ell_{c1}^2 + m_2(\ell_1^2 + \ell_{c2}^2 + 2\ell_1 \ell_{c2} \cos \theta_2) + I_1 + I_2 \quad (1.260)$$

$$M_{12} = m_2(\ell_{c2}^2 + \ell_1 \ell_{c2} \cos \theta_2) + I_2 \quad (1.261)$$

$$M_{22} = m_2 \ell_{c2}^2 + I_2 \quad (1.262)$$

1.16.3 CNC Machine Tools

Axis Drive Model

Each axis of a CNC machine:

$$J_{\text{axis}} \ddot{x} + B\dot{x} + F_c \operatorname{sgn}(\dot{x}) = F_{\text{motor}} - F_{\text{cut}} \quad (1.263)$$

where F_c is Coulomb friction and F_{cut} is cutting force disturbance.

Servo System

Position loop transfer function:

$$\frac{X(s)}{X_{\text{cmd}}(s)} = \frac{K_p K_v / s}{1 + K_p K_v / s + s^2 / \omega_n^2} \approx \frac{K_p}{\tau_p s + 1} \quad (1.264)$$

for well-tuned systems where $K_v = K_p / \tau_p$.

1.16.4 Power Systems

Synchronous Generator

Swing equation:

$$\frac{2H}{\omega_s} \frac{d^2 \delta}{dt^2} + D \frac{d\delta}{dt} = P_m - P_e \quad (1.265)$$

where:

- H = inertia constant (seconds)
- δ = rotor angle (electrical radians)
- ω_s = synchronous speed
- P_m = mechanical power
- P_e = electrical power

Linearized about operating point:

$$\frac{\Delta \delta(s)}{\Delta P_m(s)} = \frac{\omega_s / 2H}{s^2 + (D\omega_s / 2H)s + K_s \omega_s / 2H} \quad (1.266)$$

where $K_s = \partial P_e / \partial \delta$ is the synchronizing coefficient.

Automatic Generation Control (AGC)

Area control error:

$$ACE = \Delta P_{tie} + B\Delta f \quad (1.267)$$

where B is the frequency bias setting.

1.16.5 Queueing Systems

M/M/1 Queue

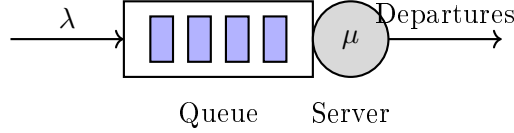


Figure 1.50: M/M/1 queue model.

Steady-state:

$$\text{Utilization: } \rho = \lambda/\mu \quad (1.268)$$

$$\text{Mean queue length: } L = \frac{\rho}{1 - \rho} \quad (1.269)$$

$$\text{Mean waiting time: } W = \frac{1}{\mu - \lambda} \quad (1.270)$$

Fluid Flow Model

For control design, queues can be approximated as fluid:

$$\frac{dq}{dt} = \lambda(t) - \mu(t) \quad (1.271)$$

where q is queue length, λ is arrival rate, and μ is service rate.

Transfer function:

$$\frac{Q(s)}{\Lambda(s) - M(s)} = \frac{1}{s} \quad (1.272)$$

1.16.6 Computer Systems

CPU Load Control

CPU utilization dynamics:

$$\tau \frac{dU}{dt} + U = K_u \cdot R \quad (1.273)$$

where U is utilization, R is admission rate, and τ depends on job service time.

Network Congestion Control

TCP congestion window dynamics (simplified):

$$\frac{dW}{dt} = \frac{1}{RTT} - \frac{W}{2} \cdot p \quad (1.274)$$

where W is congestion window, RTT is round-trip time, and p is packet loss probability.

Memory Management

Page fault rate model:

$$f = f_0 \left(\frac{W_s}{M} \right)^\alpha \quad (1.275)$$

where M is allocated memory, W_s is working set size, and α is a workload parameter.

1.16.7 Data Center Thermal Management

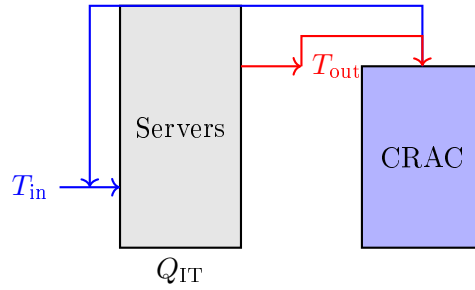


Figure 1.51: Data center thermal model.

Thermal dynamics:

$$C_{room} \frac{dT}{dt} = Q_{IT} - \dot{m}c_p(T - T_{supply}) \quad (1.276)$$

where $\dot{m}$ is air mass flow rate (controlled by CRAC fans).

1.16.8 Building Automation

HVAC Zone Model

$$C_z \frac{dT_z}{dt} = Q_{HVAC} + Q_{internal} + Q_{solar} - UA(T_z - T_{out}) \quad (1.277)$$

Transfer function from HVAC power to zone temperature:

$$\frac{T_z(s)}{Q_{HVAC}(s)} = \frac{1/UA}{\tau s + 1} \quad (1.278)$$

where $\tau = C_z/UA$.

Lighting Control

Dimming dynamics:

$$\tau_L \frac{dL}{dt} + L = K_L \cdot d \quad (1.279)$$

where L is light level, d is dimmer setting, and τ_L is lamp response time.

1.16.9 Traffic Flow

Cell Transmission Model

For a road segment:

$$\frac{dn_i}{dt} = q_{i-1} - q_i \quad (1.280)$$

where n_i is vehicle count in cell i and q_i is flow rate.

Flow-density relationship:

$$q = v_f n \left(1 - \frac{n}{n_{\max}} \right) \quad (1.281)$$

Traffic Signal Timing

Green time allocation affects queue discharge:

$$\frac{dq}{dt} = \lambda - s \cdot g(t) \quad (1.282)$$

where s is saturation flow rate and $g(t) \in \{0, 1\}$ indicates green signal.

1.16.10 Exercises: Industrial and Computer Systems

Exercise 1.31. A conveyor belt has equivalent inertia $J = 2 \text{ kg}\cdot\text{m}^2$, friction $B = 0.5 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$, drive roller radius $r = 0.1 \text{ m}$, and motor constant $K_t = 0.5 \text{ N}\cdot\text{m}/\text{A}$.

- (a) Find the transfer function from motor current to belt velocity
- (b) Calculate the current needed for 0.5 m/s belt speed
- (c) Find the time constant

Exercise 1.32. A single-link robot arm has $J = 0.5 \text{ kg}\cdot\text{m}^2$, $B = 0.1 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$, $m = 2 \text{ kg}$, and $\ell = 0.5 \text{ m}$.

- (a) Write the nonlinear equation of motion
- (b) Linearize about $\theta_0 = 0$ (horizontal)
- (c) Find the gravity compensation torque
- (d) Derive the transfer function from torque to angle (with gravity compensated)

Exercise 1.33. An M/M/1 queue has arrival rate $\lambda = 80 \text{ jobs/s}$ and service rate $\mu = 100 \text{ jobs/s}$.

- (a) Calculate the utilization
- (b) Find the mean queue length
- (c) Find the mean response time (waiting + service)
- (d) If the arrival rate increases to 90 jobs/s, how do the metrics change?

Exercise 1.34. A data center has thermal capacitance $C = 10^6 \text{ J/K}$, heat load $Q_{IT} = 100 \text{ kW}$, and air flow rate that removes heat according to $\dot{Q}_{\text{cool}} = 5000(T - 18) \text{ W}$ where T is in $^\circ\text{C}$.

- (a) Find the steady-state temperature
- (b) Calculate the thermal time constant
- (c) If IT load suddenly increases to 120 kW, find the new steady-state and time to reach it

Exercise 1.35. A traffic intersection has arrival rate 600 vehicles/hour and saturation flow 1800 vehicles/hour. The cycle length is 90 seconds.

- (a) Find the minimum green time to prevent queue buildup
- (b) If green time is 40 seconds, calculate the average queue at the start of green
- (c) Model the queue dynamics as a hybrid system